

Best Garbling is No Garbling: Persuasion in Real Time

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Abstract

We study continuous-time persuasion where a sender controls both how informative a signal is over time and when to stop providing information to a receiver. Given an exogenous signal process, the sender can both garble the evolving signal path and delay the receiver’s decision at a convex, increasing cost of time. We show that, although both instruments are available, under natural regularity conditions, *full transparency is always optimal*: the sender keeps the signal fully informative and persuades solely by choosing when to stop.

1 Introduction

Many environments—drug approvals, loan underwriting, jury verdicts, hiring and tenure votes, content moderation, even “buy/not-buy” choices—share the same essential features: one agent assembles and releases evidence to sway another agent into taking a favorable binary decision, which takes a cut-off form. For instance, a

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regulator or internal evaluator approves a drug being tested if and only if her belief that the drug works exceeds a cutoff; a capital-provider funds a loan if and only if default risk falls below a hurdle; and a median juror votes to convict if and only if the guilt probability passes a “reasonable doubt” or “preponderance of evidence” threshold.

The canonical persuasion problem (Kamenica and Gentzkow, 2011) captures this logic in one sentence: a sender, facing a binary state and a receiver with a threshold rule for a binary action, commits to an information structure to maximize the chance the receiver’s posterior exceeds the threshold. What the classic model leaves out, and what real cases make unavoidable, is that moving beliefs takes time: evidence must be generated, filtered, and digested, and delay is costly.

In our setting, we keep this threshold-persuasion objective but make the information flow and the stopping time explicit—signals arrive over time, the receiver updates continuously and acts at the first crossing of the threshold, and the sender pays a convex cost of delay—so the design problem becomes: how should a sender shape and pace what is revealed to persuade a receiver *at the lowest (expected) cost*?

More concretely, the sender and the receiver share a common prior about an unknown binary state that drives a Brownian process X in continuous time. This process is not observed directly, but instead through a sender-chosen information structure that garbles X . The sender chooses both the informativeness of the real-time disclosure and a stopping time at which the receiver acts, incurring a convex increasing cost of delay. Upon stopping, the receiver takes a binary action, choosing the sender-preferred action if and only if her belief exceeds a threshold.

Our main result is stark: for a broad class of policies,¹ the “best garbling is no garbling.” That is, our main result, [Theorem 3.7](#) reveals that there is an optimal

¹The sender has neither advance information about future evidence nor any state information beyond the evidence already observed ([Assumption 2.1](#)). In addition, the policy is clock-dominated: at every calendar date, the receiver’s posterior equals the value reached by a copy of the transparent posterior after no more than the elapsed calendar time. This restriction is cumulative, so the sender may remain silent, later catch up, or release stored information in a lump; see [Definition 3.6](#).

solution to the sender’s persuasion problem in which she does not garble the exogenous process at all and influences the receiver’s behavior through the choice of stopping time alone.² The basic intuition for this is that garbling has a net *time-risk-increasing* effect on the receiver’s belief process, which is suboptimal given the convex time cost.

To elaborate, holding fixed the terminal posterior distribution, no public policy can produce it in less expected time than full transparency, and every transparent stopping rule that produces it has the same expected duration ([Proposition 3.2](#) and [corollary 3.3](#)). Expected duration alone does not settle the problem under a convex cost, however. Among stopping rules that produce the same terminal posterior distribution, we use a classic construction by [Root \(1969\)](#), whose barrier preserves the mean and weakly reduces the dispersion of the stopping time ([Proposition 3.5](#)).

We also study an extension where the receiver chooses when to stop, matching only the terminal posterior distribution is not enough because the receiver may prefer to stop earlier. However, under a slightly stronger (Lipschitz) condition, every equilibrium involves no garbling. The sender weakly benefits, while the receiver’s payoff rises by exactly the expected waiting cost saved ([Theorem 4.6](#)).

Finally, we study the transparent stopping problem. Higher incremental delay costs lower the corresponding minimum and maximum persuasion probabilities ([Proposition 5.5](#)). In turn, a higher signal-to-noise ratio raises the persuasion probabilities ([Proposition 5.6](#)).

[§2](#) presents the environment. [§3](#) develops the disclosure-clock comparison and proves the main theorem. [§4](#) studies receiver stopping. [§5](#) establishes the comparative statics. All proofs are collected in the appendix.

²In fact, [Theorem 3.7](#) reveals that no garbling is the fastest (thus, best) way of embedding any target distribution.

1.1 Related Work

A growing literature studies dynamic persuasion and information design. Relative to this work and the static benchmark (Kamenica and Gentzkow, 2011), our contribution is to make the tempo of information part of the designer’s choice. Technically, we bring Brownian methods into persuasion by allowing the sender to choose both how transparent to be about a continuous-time diffusion and when to stop.

Several papers study dynamic persuasion with costly or gradual information, but treat time differently. Che et al. (2023) analyze dynamic persuasion when information takes time to generate and neither player can commit. As persuasion costs vanish, Markov equilibria approximate both the Kamenica and Gentzkow (2011) optimum and full revelation, but the sender has no separate stopping instrument. Escudé and Sinander (2023) study “slow persuasion” under a gradual-accumulation constraint and characterize when the sender benefits from slowing down learning.³

Our continuous-time (Brownian) setting is related to Aybas and Callander (2024), who use Brownian motion to represent complexity and study when one-shot cheap talk can be efficient, and to work comparing cheap talk and persuasion, such as Kamenica and Lin (2024). On the static side, papers that exploit the geometric structure of persuasion problems include Bardhi and Guo (2018) and Guo and Shmaya (2019).

Closest in spirit are models where influence comes from controlling the continuation of an evidence stream. Brocas and Carrillo (2007) study a discrete-time setting in which a sender gains influence by deciding when to stop generating public evidence (“influence through ignorance”). Henry and Ottaviani (2019) explore a continuous-time approval environment in which an informer pays a flow cost to run a fixed Brownian experiment and an evaluator decides whether to approve or reject. Starting from Wald’s sequential test (Wald, 1945), they analyze how organizational form distorts stopping. Our paper tackles a different design problem: rather than fix-

³Other recent work combines dynamic disclosure with additional strategic forces like multiple audiences or incentive provision. See, e.g., Li et al. (2025) and Ely et al. (2025).

ing the experiment and varying who controls stopping, we take the continuous-time information technology itself as an object of design, allowing the sender to choose transparency as well as when to stop.⁴

Another strand combines persuasion with search and outside information sources. [Bizzotto et al. \(2021\)](#) study dynamic persuasion when the receiver also observes outside news and can stop listening, while [Mekonnen et al. \(2025\)](#) and [Mekonnen and Pakzad-Hurson \(2025\)](#) analyze information brokers who design signals for a searching agent.

On the information-acquisition side, [Zhong \(2022\)](#) and [Chen and Zhong \(2025\)](#) study single-agent dynamic problems in which a decision maker chooses a signal process subject to constraints or costs and trades off delay against accuracy, while [Bloedel and Zhong \(2020\)](#) characterize which posterior-separable reduced-form costs arise from optimal sequential sampling. [Morris and Strack \(2019\)](#) show that, in the two-state Wald problem, any distribution of posteriors can be implemented by a stopping rule and the induced ex ante cost is posterior-separable. We use a similar embedding perspective, but focus on the sender’s dynamic persuasion problem rather than on foundations of information costs.⁵

2 Model

There is an unknown state, $\mu \in \{\mu_h, \mu_\ell\}$, where $\mu_h > \mu_\ell$. A principal and an agent share a full-support prior $p_0 := \mathbb{P}(\mu = \mu_h) \in (0, 1)$. The unknown state is the drift of an evidence process X , which evolves according to

$$dX_t = \mu dt + \sigma dB_t,$$

⁴[McClellan \(2022\)](#) pushes the [Henry and Ottaviani](#) agenda further.

⁵For related continuous-time sequential-sampling models of response times and learning in rich environments, see, e.g., [Fudenberg et al. \(2018\)](#); [Gonçalves \(2023, 2024\)](#); [Barilla \(2025\)](#); [Georgiadis-Harris \(2023\)](#).

where $\sigma > 0$ is known and B is a Brownian motion independent of μ . The ratio $\kappa := (\mu_h - \mu_\ell)/\sigma$ measures the signal-to-noise ratio. Let $\mathbb{F}^X = (\mathcal{F}_t^X)_{t \geq 0}$ be the usual augmentation of the natural filtration of X .⁶

The sender may randomize her reports, and a report may contain noise that is independent of the state. Accordingly, we allow the ambient filtration $\mathbb{F}$ to contain auxiliary randomness in addition to the evidence history $\mathbb{F}^X$. Our next assumption ensures that this additional randomness neither reveals future evidence nor supplies another signal about the state.

Assumption 2.1. Let $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ be a usual filtration containing $\mathbb{F}^X$, and let $\mathbb{P}^\ell$ and $\mathbb{P}^h$ be the conditional laws under the low and high states. Under $\mathbb{P}^\ell$, assume $B_t^\ell := (X_t - \mu_\ell t)/\sigma$ is an $\mathbb{F}$ -Brownian motion. For every finite t , assume

$$\left. \frac{d\mathbb{P}^h}{d\mathbb{P}^\ell} \right|_{\mathcal{F}_t} = Z_t := \mathcal{E}(\kappa B^\ell)_t = \exp \left(\kappa B_t^\ell - \frac{\kappa^2}{2} t \right).$$

Assumption 2.1 imposes two restrictions on the information in $\mathbb{F}$. First, requiring B^ℓ to remain an $\mathbb{F}$ -Brownian motion prevents $\mathcal{F}_t$ from revealing future increments of the evidence process. Second, the likelihood-ratio identity says that the evidence history contains all of the state information in $\mathcal{F}_t$. In particular, for every $A \in \mathcal{F}_t$, $\mathbb{P}^h(A | \mathcal{F}_t^X) = \mathbb{P}^\ell(A | \mathcal{F}_t^X)$. Thus, the sender may use the additional randomness in $\mathbb{F}$ to randomize or add noise to her reports, but that randomness cannot reveal anything about the state once the evidence history is known.

Girsanov's theorem also produces the evidence dynamics under the high state. Under $\mathbb{P}^h$, $B_t^h := B_t^\ell - \kappa t$ is an $\mathbb{F}$ -Brownian motion, and therefore $X_t = \mu_h t + \sigma B_t^h$. Consequently, observing all of $\mathcal{F}_t$ produces the same posterior as observing the evidence history alone:

$$\mathbb{P}(\mu = \mu_h | \mathcal{F}_t) = \mathbb{P}(\mu = \mu_h | \mathcal{F}_t^X) = \frac{p_0 Z_t}{1 - p_0 + p_0 Z_t}.$$

⁶The natural filtration records the history of the process. A filtration is *usual* if it is complete and right-continuous; its usual augmentation adds null events and imposes right-continuity.

Because $\mathbb{P}^h$ and $\mathbb{P}^\ell$ are the laws conditional on the high and low states, the law of total probability yields, for every event A , $\mathbb{P}(A) = p_0 \mathbb{P}^h(A) + (1 - p_0) \mathbb{P}^\ell(A)$.

Microfounding the Reporting environment

The following product construction provides a concrete microfoundation for [Assumption 2.1](#). Let $(\Omega^r, \mathcal{R}, \mathbb{Q})$ carry a random seed ξ ,⁷ an m -dimensional Brownian motion V , and a Poisson random measure Π on $\mathbb{R}_+ \times E$ with deterministic compensator $dt \varrho(dz)$, where E is a standard Borel space and ϱ is σ -finite. Assume that ξ , V , and Π are mutually independent under $\mathbb{Q}$, and let $\mathbb{U} = (\mathcal{U}_t)_{t \geq 0}$ be the $\mathbb{Q}$ -usual augmentation of the filtration they generate, with $\xi \in \mathcal{U}_0$. Under state $i \in \{h, \ell\}$, take $\mathbb{P}^i := \mathbb{P}_X^i \otimes \mathbb{Q}$, where $\mathbb{P}_X^i$ is the state- i law of X . Thus, only the law of the evidence coordinate varies with the state.

Let $\mathbb{F}^{r,0} := (\mathcal{F}_t^{r,0})_{t \geq 0}$, where $\mathcal{F}_t^{r,0} := \mathcal{F}_t^X \vee \mathcal{U}_t$, and let $\mathbb{F}^r = (\mathcal{F}_t^r)_{t \geq 0}$ be its usual augmentation under $\mathbb{P}$. The filtration $\mathbb{F}^r$ records the evidence and the state-independent randomization available to the reporting device through each date.

The sender may commit to any d -dimensional càdlàg $\mathbb{F}^r$ -adapted report Y that admits the representation

$$Y_t = Y_0 + D_t^Y + \int_0^t \beta_s dX_s + \int_0^t \gamma_s dV_s + \int_{(0,t] \times E} \delta_s(z) \tilde{\Pi}(ds, dz), \quad (1)$$

where $\tilde{\Pi}(ds, dz) := \Pi(ds, dz) - ds \varrho(dz)$. Here, Y_0 is $\mathcal{F}_0^r$ -measurable, $D_0^Y = 0$, and D^Y is an $\mathbb{F}^r$ -adapted càdlàg process with finite variation on every bounded interval. The integrands β , γ , and δ are $\mathbb{F}^r$ -predictable and make the three stochastic integrals well defined on every bounded interval under either state. The predictable coefficients may depend on earlier evidence, reports, and auxiliary draws, while D^Y may also jump in response to information available at the jump date. Let $\mathbb{G}^Y$ denote the usual augmentation of the natural filtration of Y .

⁷An initial random variable ξ used for date-zero randomization.

Report taxonomy. The representation in (1) contains the following reporting technologies.

- (i) *Full transparency and continuous reports.* Taking $\beta_t = 1$ and setting the remaining terms equal to zero is full transparency. With no Poisson term and continuous D^Y , the same equation permits multidimensional continuous reports that transform the evidence according to the earlier history and may add independent Gaussian noise.
- (ii) *Marked and categorical messages.* A finite or countable subset of E may index message labels. The reporting rule may make the arrival rate and the distribution of labels depend on earlier evidence, so both the arrival of a message and its continued absence may be informative.
- (iii) *Silence, delayed disclosure, and batches.* If the committed rule makes D^Y constant and β , γ , and δ vanish on an interval, the report is silent there. If the duration of silence depends on earlier evidence, continued silence may itself be informative. Jumps of D^Y or of the marked component permit delayed announcements, summaries of earlier evidence, and batched disclosure.
- (iv) *Lévy-type reports.* The auxiliary Brownian and Poisson drivers permit continuous and jump noise. Time-homogeneous characteristics produce ordinary Lévy components, while predictable characteristics produce more general Itô-Lévy reports and random announcement times.
- (v) *Hybrid reports.* A single multidimensional report may combine continuous noisy disclosure, categorical alerts, silence, random report dates, and delayed or batched announcements.

Proposition 2.2. *In the product construction above, let $\tilde{\mathbb{F}} = (\tilde{\mathcal{F}}_t)_{t \geq 0}$ be any usual filtration satisfying $\mathcal{F}_t^X \subseteq \tilde{\mathcal{F}}_t \subseteq \mathcal{F}_t^r$ for every $t \geq 0$. Then $\tilde{\mathbb{F}}$ satisfies [Assumption 2.1](#). In particular, the conclusion holds for the usual augmentation of $(\mathcal{F}_t^X \vee \mathcal{G}_t^Y)_{t \geq 0}$.*

The product construction limits what the reporting device may use: all state information enters through X , while ξ , V , and Π only randomize how the device

transforms and releases that evidence.

The sender observes X in real time. The receiver observes only a committed public report and forms a posterior $\hat{P}$ from its public history. At the sender's public stopping time τ , the receiver chooses an action in $\{0, 1\}$. There is a threshold $\bar{P} \in (0, 1)$ such that the receiver takes the sender-preferred action if and only if $\hat{P}_\tau \geq \bar{P}$. The sender receives one when that action is taken and zero otherwise. Her delay cost $c: [0, \infty) \rightarrow [0, \infty)$ is finite, continuous, nondecreasing, convex, and satisfies $c(0) = 0$. Her payoff is

$$\mathbb{E} \left[\mathbb{1}_{\{\hat{P}_\tau \geq \bar{P}\}} - c(\tau) \right].$$

We restrict attention to almost surely finite policies satisfying $\mathbb{E}[\tau] < \infty$. A policy with $\mathbb{E}[c(\tau)] = \infty$ has payoff $-\infty$.

3 Disclosure Clocks and Transparent Persuasion

The main result of this section is a transparent-replacement theorem. A public experiment specifies the information revealed to both players over time. Formally, it is a usual subfiltration $\mathbb{G} = (\mathcal{G}_t)_{t \geq 0}$ of $\mathbb{F}$, where $\mathcal{G}_t$ records the public report history through date t . A public policy consists of a public experiment and an almost surely finite $\mathbb{G}$ -stopping time τ . Write $\hat{P}_t$ for the receiver's posterior at date t .

Full transparency Under full transparency, the receiver observes the entire evidence history. Let $P_t := \mathbb{P}(\mu = \mu_h \mid \mathcal{F}_t)$ be the resulting posterior. We note the following easy facts:

Remark 3.1. Maintain [Assumption 2.1](#). There is an $\mathbb{F}$ -Brownian motion W under $\mathbb{P}$ such that

$$dP_t = \kappa P_t(1 - P_t) dW_t, \quad \text{with } P_0 = p_0. \quad (1)$$

Moreover,

$$\log \frac{P_t}{1 - P_t} = \log \frac{p_0}{1 - p_0} + \frac{\mu_h - \mu_\ell}{\sigma^2} \left(X_t - \frac{\mu_h + \mu_\ell}{2} t \right). \quad (2)$$

Consequently, $P_t \in (0, 1)$ at every finite time, and the natural filtrations of P and X coincide up to completion.

(1) identifies the natural speed of transparent learning. When the posterior is p , its instantaneous volatility is $\kappa p(1 - p)$; that is, a higher signal-to-noise ratio makes beliefs move faster. Moreover, the histories of P and X contain the same information. A transparent stopping rule can consequently be written using either process.

3.1 Expected Time to Produce a Terminal Posterior Law

We now ask how much calendar time a public policy must take on average to produce a specified terminal distribution over posteriors. In particular, we derive a lower bound on $\mathbb{E}[\tau]$ for every public policy that produces the distribution. We then show that, under full transparency, every integrable stopping rule that produces the specified terminal posterior distribution attains this lower bound.

For a public experiment $\mathbb{G} = (\mathbb{G}_t)_{t \geq 0}$, define its public likelihood ratio and corresponding posterior by

$$\Lambda_t := \left. \frac{d\mathbb{P}^h}{d\mathbb{P}^\ell} \right|_{\mathbb{G}_t} \quad \text{and} \quad \hat{P}_t := \frac{p_0 \Lambda_t}{1 - p_0 + p_0 \Lambda_t}.$$

For probability measures $\mathbb{P} \ll \mathbb{Q}$, write

$$D(\mathbb{P} \parallel \mathbb{Q}) := \int \log \left(\frac{d\mathbb{P}}{d\mathbb{Q}} \right) d\mathbb{P},$$

for relative entropy, which measures how well the public history distinguishes one state from the other. Define

$$\Psi(p) := (2p - 1) \log \frac{p}{1 - p}, \quad \text{for } p \in (0, 1).$$

For a public policy that stops at τ , the quantity $\mathbb{E}[\Psi(\hat{P}_\tau)] - \Psi(p_0)$ equals p_0 times the relative entropy from the distribution of the public report history under the high state to its distribution under the low state, plus $1 - p_0$ times the relative entropy in the opposite direction. We say that a stopping time S *embeds* a probability distribution ν in a posterior process M if stopping M at S produces ν , meaning that $\mathcal{L}(M_S) = \nu$.

Proposition 3.2. *Maintain [Assumption 2.1](#). Let τ be an almost surely finite $\mathbb{G}$ -stopping time satisfying $\mathbb{E}[\tau] < \infty$, and set $Q := \hat{P}_\tau$. Then*

$$\mathbb{E}[\tau] \geq \frac{2}{\kappa^2} (\mathbb{E}[\Psi(Q)] - \Psi(p_0)).$$

Let M be any solution in a usual filtration to

$$dM_t = \kappa M_t(1 - M_t) dW_t, \quad \text{with } M_0 = p_0. \quad (2)$$

If S is a stopping time satisfying $\mathbb{E}[S] < \infty$ and $(M_{t \wedge S})_{t \geq 0}$ is uniformly integrable,⁸ then

$$\mathbb{E}[S] = \frac{2}{\kappa^2} (\mathbb{E}[\Psi(M_S)] - \Psi(p_0)).$$

We call any solution M in the second part of [Proposition 3.2](#) a *full-speed posterior*.

The full-transparency process P satisfies the stochastic differential equation for M specified in the second part of [Proposition 3.2](#). Because P is bounded, its stopped process is uniformly integrable. Thus, plugging in $M = P$ in that part of the proposition reveals that every integrable stopping time embedding ν in P has the same expectation. The first part of the proposition shows that every public policy producing ν has expected duration weakly greater than this common expectation. Formally,

Corollary 3.3. *Maintain [Assumption 2.1](#). Fix a distribution ν on $(0, 1)$ that can be embedded in P by an integrable stopping time. Every integrable stopping time that embeds ν in P has the same expectation. Moreover, if a public policy stops at τ , where $\mathbb{E}[\tau] < \infty$ and $\mathcal{L}(\hat{P}_\tau) = \nu$, then its expected duration is weakly greater than this common expectation.*

The corollary settles the comparison of means: holding the terminal posterior distribution fixed, every transparent stopping rule that produces it has the same

⁸Uniform integrability is the standard condition that permits expectations to pass to an unbounded stopping time. The posterior processes used below are bounded, so their stopped versions satisfy it automatically.

expected duration, and no public policy can produce it in less expected time. This corollary is constant-flow-cost case of [Morris and Strack \(2019, Propositions 2-3\)](#). Their identity continues to hold in our setting because the additional randomization allowed by [Assumption 2.1](#) does not change the evolution of the posterior under full transparency. The lower bound for public policies is a direct extension due to the convexity of Ψ .

3.2 Timing Risk

Expected time is not enough when delay costs are convex. Two stopping times may have the same mean but different dispersion, and the more dispersed time is more costly under every strictly convex delay cost. We use the convex and the increasing convex orders to make this precise.

Definition 3.4. For nonnegative integrable random variables S and T , write $S \preceq_{\text{icx}} T$ if $\mathbb{E}[\phi(S)] \leq \mathbb{E}[\phi(T)]$ for every increasing convex function ϕ for which the expectations exist. Write $S \preceq_{\text{cx}} T$ if the same inequality holds for every convex ϕ .⁹

[Root \(1969\)](#) introduces a stopping rule based on a deterministic barrier in the plane with time on one axis and the posterior on the other.¹⁰ For a continuous posterior process M , write $R_{\mathcal{B}}(M) := \inf\{t \geq 0: (t, M_t) \in \mathcal{B}\}$ for the first date at which the current date and posterior lie in $\mathcal{B}$.

Our next proposition combines [Cox et al. \(2019, Theorems 2.8 and 3.3\)](#) with the mean identity in [Proposition 3.2](#).

Proposition 3.5 (Root’s timing comparison). *Maintain [Assumption 2.1](#). Let M be a $[0, 1]$ -valued process adapted to a usual filtration, and suppose that M satisfies the*

⁹That is to say, $S \preceq_{\text{cx}} T$ if and only if $S \preceq_{\text{icx}} T$ and $\mathbb{E}[S] = \mathbb{E}[T]$.

¹⁰A Root barrier is a closed set $\mathcal{B} \subseteq [0, \infty) \times (0, 1)$ such that, if $(t, p) \in \mathcal{B}$, then $(s, p) \in \mathcal{B}$ for every $s \geq t$. Thus, at each posterior p , the barrier specifies a cutoff date, possibly infinite. The rule stops when the current posterior is p at or after that date. The barrier is deterministic because these cutoff dates are fixed before the process begins.

stochastic differential equation (2) with respect to a Brownian motion in that filtration. Let ν be a probability distribution on $(0, 1)$, and suppose that some integrable stopping time embeds ν in M . Then there is a deterministic Root barrier $\mathcal{B}^\nu$ such that $R_{\mathcal{B}^\nu}(M)$ is integrable, embeds ν in M , and satisfies $R_{\mathcal{B}^\nu}(M) \preceq_{\text{cx}} S$ for every integrable stopping time S that embeds ν in M .

To interpret the proposition in the present setting, take $M = P$, the posterior under full transparency. Fix a distribution ν , and suppose that an integrable stopping time S embeds ν in P . Root's barrier produces a first-entry time $R^\nu := R_{\mathcal{B}^\nu}(P)$ that also embeds ν in P . In the form stated in Lemma A.3, Cox et al. (2019) show that $R^\nu \preceq_{\text{icx}} S$. By Proposition 3.2, R^ν and S have the same mean because they embed the same terminal posterior distribution in P . Together, these two results imply $R^\nu \preceq_{\text{cx}} S$ and so, among the stopping times that embed ν in P , Root's construction preserves expected duration and weakly reduces dispersion.

To apply Proposition 3.5 to a public policy, it remains to express the policy's posterior on the operating-time scale of a full-speed posterior.

3.3 Clock Domination

A public report determines a posterior process $\hat{P}$. We compare its timing with full transparency by writing $\hat{P}_t = Q_{C_t}$, where Q satisfies (2). The clock C_t measures how long Q takes to reach the public posterior at date t .

Definition 3.6. Let $\mathbb{G}$ be a public experiment. A *disclosure clock* for $\mathbb{G}$ is an increasing right-continuous $\mathbb{G}$ -adapted process $C = (C_t)_{t \geq 0}$, with $C_0 = 0$, for which, after enlarging the probability space by independent auxiliary randomness if necessary, there exist a usual filtration $\mathbb{J} = (\mathcal{G}_u)_{u \geq 0}$ and a $\mathbb{J}$ -adapted process Q satisfying (2) with respect to a $\mathbb{J}$ -Brownian motion such that:

- (i) each C_t is a finite $\mathbb{J}$ -stopping time;
- (ii) $\mathcal{G}_t \subseteq \mathcal{G}_{C_t}$ and $\hat{P}_t = Q_{C_t}$ for every $t \geq 0$; and

(iii) C_τ is a $\mathbb{J}$ -stopping time for every public stopping rule τ .

The public experiment is *clock-dominated* if it has a disclosure clock C satisfying $C_t \leq t$ for every $t \geq 0$ almost surely. A *clock-dominated public policy* is a triple $(\mathbb{G}, C, \tau)$, where C is such a clock and τ is an integrable public stopping rule.

If $C_t = s$, then the public posterior at date t equals the posterior that a copy of full transparency reaches after s units of time. Silence makes C flat, faster disclosure makes it steep, and releasing a stored interval makes it jump. Clock domination allows catch-up and jumps but requires $C_t \leq t$ at every date.

The stopping-time requirements in [Definition 3.6](#) ensure that a public stopping rule τ becomes a stopping time C_τ for Q . [Lemma A.4](#) constructs a disclosure clock when the public likelihood is continuous, and [Proposition A.14](#) constructs one for delayed or batched release of a stored record.

3.4 Best Garbling Is No Garbling

Let $\mathbb{F}^P$ be the usual augmentation of the natural filtration of P . Fix a bounded measurable terminal payoff $F: (0, 1) \rightarrow \mathbb{R}$ and an increasing convex delay cost $c: [0, \infty) \rightarrow \mathbb{R}$. Define

$$V_{F,c}^{\text{garb}} := \sup_{(\mathbb{G}, C, \tau) \text{ a clock-dominated public policy}} \mathbb{E}[F(\hat{P}_\tau) - c(\tau)]$$

and

$$V_{F,c}^0 := \sup_{\substack{\rho \text{ an } \mathbb{F}^P\text{-stopping time} \\ \mathbb{E}[\rho] < \infty}} \mathbb{E}[F(P_\rho) - c(\rho)].$$

Because F is bounded and c is bounded below, both values are finite.

Theorem 3.7. *Maintain [Assumption 2.1](#). Let $(\mathbb{G}, C, \tau)$ be a clock-dominated public policy, and set $\nu := \mathcal{L}(\hat{P}_\tau)$. There is an integrable $\mathbb{F}^P$ -stopping time R^ν such that $\mathcal{L}(P_{R^\nu}) = \nu$, $R^\nu \preceq_{\text{cx}} C_\tau$, and $C_\tau \leq \tau$ almost surely; and so $R^\nu \preceq_{\text{icx}} \tau$.*

The terminal posterior distributions agree. By [Definition 3.4](#), $R^\nu \preceq_{\text{cx}} C_\tau$ means that R^ν and C_τ have the same mean and that R^ν is weakly less dispersed. Clock domination implies $C_\tau \leq \tau$; consequently, $\mathbb{E}[F(P_{R^\nu}) - c(R^\nu)] \geq \mathbb{E}[F(\hat{P}_\tau) - c(\tau)]$.

Full transparency is clock-dominated with $C_t = t$. Hence, [Theorem 3.7](#) implies $V_{F,c}^{\text{garb}} = V_{F,c}^0$. The proof of [Theorem 3.7](#) also produces:

- (i) if $\tau \leq T$ almost surely for some deterministic $T < \infty$, then $R^\nu \leq T$ almost surely;
- (ii) if $\mathbb{E}[c(\tau)] < \infty$, c is strictly increasing, and $\mathbb{P}(C_\tau < \tau) > 0$, then $\mathbb{E}[c(R^\nu)] < \mathbb{E}[c(\tau)]$; and
- (iii) if $\mathbb{E}[c(C_\tau)] < \infty$, c is strictly convex, and $\mathcal{L}(R^\nu) \neq \mathcal{L}(C_\tau)$, then $\mathbb{E}[c(R^\nu)] < \mathbb{E}[c(C_\tau)]$.

For the binary-action persuasion problem in [§2](#), take $F(p) := \mathbb{1}_{\{p \geq \bar{P}\}}$.

4 An Uncommitted Receiver

The preceding results let the sender choose when the experiment ends. We now let the receiver choose when to stop after the sender commits to a clock-dominated public experiment with a fixed disclosure clock. Retain the binary terminal action from [§2](#). At posterior p , action $a \in \{0, 1\}$ delivers the receiver expected payoff $u_R(p, a)$, which is affine in p , and action one is optimal if and only if $p \geq \bar{P}$. Define $r(p) := \max_{a \in \{0, 1\}} u_R(p, a)$, which is bounded, continuous, and convex. The receiver incurs the linear waiting cost $\kappa_R t$, where $\kappa_R > 0$.

Definition 4.1. A receiver equilibrium is a pair (S, ρ) , where $S = (\mathbb{G}^S, C)$, $\mathbb{G}^S$ is a public experiment, and C is a fixed disclosure clock for $\mathbb{G}^S$ satisfying, almost surely, $C_t \leq t$ for every $t \geq 0$. Write $\hat{P}^S$ for the public posterior under $\mathbb{G}^S$. The stopping time ρ is a $\mathbb{G}^S$ -stopping time attaining

$$\sup_{\substack{\tau \text{ a } \mathbb{G}^S\text{-stopping time} \\ \mathbb{E}[\tau] < \infty}} \mathbb{E}[r(\hat{P}_\tau^S) - \kappa_R \tau].$$

Definition 4.2. A disclosure clock C is unbanked if, almost surely, $C_t - C_s \leq t - s$ for every $0 \leq s \leq t$.

An unbanked clock is continuous and satisfies $C_t \leq t$. It permits silence but not later disclosure above full speed. This stronger restriction matters because the receiver may stop before the experiment ends.

Fix a receiver equilibrium (S, ρ) with an unbanked disclosure clock C . Because C is a disclosure clock, there exist a usual filtration $\mathbb{J}$ and a $\mathbb{J}$ -adapted process Q satisfying (2) and the requirements in Definition 3.6. Fix such $\mathbb{J}$ and Q , and set $U := C_\rho$. Then U is a $\mathbb{J}$ -stopping time and $\hat{P}_\rho^S = Q_U$.

We first relate stopping Q before U to stopping the original experiment in calendar time. For $u \geq 0$, define $T_u^+ := \inf\{t \geq 0: C_t > u\}$ and $\theta_u := \rho \wedge T_u^+$, where $\inf \emptyset = \infty$. The date θ_u is the earlier of receiver stopping and the calendar date at which the disclosure clock first moves strictly above u . The family $(\theta_u)_{u \geq 0}$ is increasing and right-continuous.

Define $\mathcal{H}_u^0 := \mathcal{G}_{\theta_u}^S$, and write $\mathbb{H}^0 = (\mathcal{H}_u^0)_{u \geq 0}$. If $u_n \downarrow u$, then $\theta_{u_n} \downarrow \theta_u$, and right-continuity of $\mathcal{G}^S$ implies $\bigcap_n \mathcal{H}_{u_n}^0 = \mathcal{H}_u^0$. Hence, $\mathbb{H}^0$ is right-continuous. Let $\mathbb{H} = (\mathcal{H}_u)_{u \geq 0}$ be its completion. The strict inequality in the definition of T_u^+ selects the end of a flat stretch of the clock. This retains any state-independent public information revealed while the clock is paused.

Lemma 4.3. *Maintain Assumption 2.1. Let (S, ρ) be a receiver equilibrium whose fixed clock C is unbanked. Then U is an $\mathbb{H}$ -stopping time. For every $u \geq 0$, $C_{\theta_u} = u \wedge U$ and $\hat{P}_{\theta_u}^S = Q_{u \wedge U}$. If $V \leq U$ is an $\mathbb{H}$ -stopping time, then θ_V is a $\mathcal{G}^S$ -stopping time, $\hat{P}_{\theta_V}^S = Q_V$, and $\theta_V - V \leq \rho - U$ almost surely.*

If the receiver would stop Q at $V \leq U$, she can instead stop the original experiment at θ_V . The two rules produce the same posterior. Moreover, $\theta_V - V \leq \rho - U$, so the alternative rule adds no more calendar delay than the equilibrium rule itself adds.

Let $\mathbb{F}^Q$ denote the usual augmentation of the natural filtration of Q . We permit a stopping rule to use an independent random variable drawn at date zero. This variable affects only the stopping date and is not separately disclosed to the receiver.

Lemma 4.4. *Maintain [Assumption 2.1](#). Retain Q and U from [Lemma 4.3](#). There is a stopping time σ_D for the filtration generated by P and that variable such that $((P_u)_{u \geq 0}, \sigma_D) \stackrel{d}{=} ((Q_u)_{u \geq 0}, U)$. Define*

$$Z^D := \left((P_{u \wedge \sigma_D})_{u \geq 0}, (\mathbb{1}_{\{\sigma_D \leq u\}})_{u \geq 0} \right) \quad \text{and} \quad Z^Q := \left((Q_{u \wedge U})_{u \geq 0}, (\mathbb{1}_{\{U \leq u\}})_{u \geq 0} \right).$$

Then $Z^D \stackrel{d}{=} Z^Q$. Every stopping time $\eta \leq \sigma_D$ for the usual natural filtration of Z^D has a stopping time $V \leq U$ for the usual natural filtration of Z^Q such that $(P_\eta, \eta) \stackrel{d}{=} (Q_V, V)$.

By [Lemma 4.4](#), each stopping rule $\eta \leq \sigma_D$ under transparency has a stopping rule $V \leq U$ satisfying $(P_\eta, \eta) \stackrel{d}{=} (Q_V, V)$. By [Lemma 4.3](#), θ_V is a stopping rule under $\mathbb{S}$, produces posterior Q_V , and satisfies $\theta_V - V \leq \rho - U$.

Definition 4.5. The transparent acceleration D reveals the fully informed posterior P until σ_D , publicly announces termination at σ_D , and reveals no further information afterward. Its public filtration $\mathbb{G}^D = (\mathbb{G}_t^D)_{t \geq 0}$ is the usual augmentation of $\mathbb{G}^{D,0} = (\mathbb{G}_t^{D,0})_{t \geq 0}$, where

$$\mathbb{G}_t^{D,0} := \sigma \left(P_{s \wedge \sigma_D}, \mathbb{1}_{\{\sigma_D \leq s\}} : 0 \leq s \leq t \right).$$

The transparent acceleration has disclosure clock $C_t^D = t \wedge \sigma_D$. Thus, it reveals at full speed until σ_D and then stops permanently. The verification is included in the proof of [Theorem 4.6](#).

Theorem 4.6. *Maintain [Assumption 2.1](#). Let $(\mathbb{S}, \rho)$ be a receiver equilibrium with unbanked disclosure clock C . Then σ_D is a receiver best response to D . The sender's payoff satisfies $\mathbb{E} \left[\mathbb{1}_{\{P_{\sigma_D} \geq \bar{P}\}} - c(\sigma_D) \right] \geq \mathbb{E} \left[\mathbb{1}_{\{\hat{P}_\rho^S \geq \bar{P}\}} - c(\rho) \right]$, and the receiver's payoff satisfies $\mathbb{E}[r(P_{\sigma_D}) - \kappa_R \sigma_D] = \mathbb{E}[r(\hat{P}_\rho^S) - \kappa_R \rho] + \kappa_R \mathbb{E}[\rho - C_\rho]$.*

By Lemma 4.4, P_{σ_D} and $\hat{P}_\rho^S$ have the same distribution, while σ_D and C_ρ have the same distribution. Since $C_\rho \leq \rho$, the sender's terminal payoff is unchanged and her expected delay cost weakly falls. The receiver's terminal value is also unchanged, while her expected waiting cost falls by exactly $\kappa_R \mathbb{E}[\rho - C_\rho]$.

Corollary 4.7. *Maintain Assumption 2.1. Every receiver equilibrium whose fixed clock is unbanked has a transparent acceleration that remains optimal for the receiver and weakly benefits both players.*

5 Comparative Statics

The preceding results reduce optimization within the clock-dominated class to transparent stopping. Let $\mathbb{F}^P$ be the usual augmentation of the natural filtration of P , so that transparent stopping rules are precisely the $\mathbb{F}^P$ -stopping times, and define

$$H := \inf\{t \geq 0: P_t = \bar{P}\} \quad \text{and} \quad \mathcal{T}_H := \left\{ \rho: \rho \text{ is an } \mathbb{F}^P\text{-stopping time, } \rho \leq H \text{ a.s., } \mathbb{E}[\rho] < \infty \right\}.$$

For a cost c , define

$$J_c(\rho) := \mathbb{P}(\rho = H) - \mathbb{E}[c(\rho)], \quad \mathcal{V}(c) := \sup_{\rho \in \mathcal{T}_H} J_c(\rho), \quad \text{and} \quad \mathcal{S}(c) := \arg \max_{\rho \in \mathcal{T}_H} J_c(\rho).$$

The objective $J_c(\rho)$ is well defined in $[-\infty, 1]$, and immediate stopping delivers value zero.

Lemma 5.1. *Maintain Assumption 2.1. Every integrable $\mathbb{F}^P$ -stopping time ρ is dominated by $\rho \wedge H$. If ρ is optimal for the transparent problem, then $\rho \wedge H$ is also optimal and has the same persuasion probability.*

Lemma 5.2. *Maintain Assumption 2.1. For every delay cost c , the set $\mathcal{S}(c)$ is nonempty, is closed under pairwise minimum and maximum, and contains a least and a greatest element under the order $\rho \leq \eta$ almost surely.*

The first lemma says that an optimal transparent rule never waits after the posterior has reached the persuasion threshold. The second says that optimal normalized rules have an ordered structure: taking the earlier or later of two optimal rules remains optimal, and there are well-defined earliest and latest optimal rules. Together, the lemmas imply that every element of $\mathcal{S}(c)$ is optimal for the unrestricted transparent problem and that every transparent optimum has a representative in $\mathcal{S}(c)$ with the same persuasion probability.

By [Lemma 5.2](#), write ρ_c^- and ρ_c^+ for the least and greatest elements of $\mathcal{S}(c)$ under this order. Because $\{\rho = H\} \subseteq \{\eta = H\}$ whenever $\rho \leq \eta \leq H$, the minimum and maximum persuasion probabilities attained by optimal transparent stopping rules are $\underline{q}(c) := \mathbb{P}(\rho_c^- = H)$ and $\bar{q}(c) := \mathbb{P}(\rho_c^+ = H)$, respectively.

Definition 5.3. For nonempty $\mathcal{A}, \mathcal{B} \subseteq \mathcal{I}_H$, write $\mathcal{A} \succeq_{\text{SSO}} \mathcal{B}$ if, for every $\rho \in \mathcal{A}$ and $\eta \in \mathcal{B}$, $\rho \vee \eta \in \mathcal{A}$ and $\rho \wedge \eta \in \mathcal{B}$.

5.1 Delay Costs

Definition 5.4. A cost c_2 has higher incremental delay costs than c_1 if $t \mapsto c_2(t) - c_1(t)$ is nondecreasing.

Proposition 5.5. *Maintain [Assumption 2.1](#). If c_2 has higher incremental delay costs than c_1 , then $\mathcal{S}(c_1) \succeq_{\text{SSO}} \mathcal{S}(c_2)$. Consequently, $\rho_{c_2}^- \leq \rho_{c_1}^-$, $\rho_{c_2}^+ \leq \rho_{c_1}^+$, $\underline{q}(c_2) \leq \underline{q}(c_1)$, and $\bar{q}(c_2) \leq \bar{q}(c_1)$*

Thus, making additional delay more costly shifts both the earliest and the latest optimal stopping rules weakly earlier. Because reaching H is more likely under a later rule, both the minimum and maximum persuasion probabilities attained by optimal policies weakly fall.

5.2 Signal-to-Noise Ratio

To compare different signal-to-noise ratios on a common process, let P^1 satisfy

$$dP_u^1 = P_u^1(1 - P_u^1) dW_u, \quad \text{with } P_0^1 = p_0,$$

and let $\mathbb{F}^1 = (\mathcal{F}_u^1)_{u \geq 0}$ be the usual augmentation of its natural filtration. Define $H_1 := \inf\{u \geq 0: P_u^1 = \bar{P}\}$, and let $\mathcal{T}_{H_1}^1$ be the $\mathbb{F}^1$ -stopping times U satisfying $U \leq H_1$ almost surely and $\mathbb{E}[U] < \infty$. For each $\kappa > 0$, define

$$d_\kappa(u) := c\left(\frac{u}{\kappa^2}\right), \quad J_\kappa^1(U) := \mathbb{P}(U = H_1) - \mathbb{E}[d_\kappa(U)], \quad \text{and} \quad \mathcal{S}^1(\kappa) := \arg \max_{U \in \mathcal{T}_{H_1}^1} J_\kappa^1(U).$$

Applying [Lemma 5.2](#) to this common process produces least and greatest elements U_κ^- and U_κ^+ . Define

$$\underline{q}_{\text{snr}}(\kappa) := \mathbb{P}(U_\kappa^- = H_1) \quad \text{and} \quad \bar{q}_{\text{snr}}(\kappa) := \mathbb{P}(U_\kappa^+ = H_1).$$

For each $\kappa > 0$, set $P_t^\kappa := P_{\kappa^2 t}^1$. Then $W_t^\kappa := \kappa^{-1} W_{\kappa^2 t}$ is a Brownian motion and

$$dP_t^\kappa = \kappa P_t^\kappa(1 - P_t^\kappa) dW_t^\kappa.$$

Moreover, $\mathcal{F}_t^{P^\kappa} = \mathcal{F}_{\kappa^2 t}^1$. Thus, every signal-to-noise ratio is a deterministic time change of the same posterior process. The map $\rho \mapsto U = \kappa^2 \rho$ is a bijection between integrable stopping times for P^κ and integrable stopping times for P^1 . It maps the first hitting time of $\bar{P}$ under P^κ to H_1 , preserves the persuasion event, and transforms calendar cost $c(\rho)$ into information-time cost $d_\kappa(U)$. A higher κ makes each unit of information time require less calendar time.

Proposition 5.6. *If $\kappa_2 \geq \kappa_1$, then $\mathcal{S}^1(\kappa_2) \succeq_{\text{SSO}} \mathcal{S}^1(\kappa_1)$. Consequently, $U_{\kappa_1}^- \leq U_{\kappa_2}^-$, $U_{\kappa_1}^+ \leq U_{\kappa_2}^+$, $\underline{q}_{\text{snr}}(\kappa_1) \leq \underline{q}_{\text{snr}}(\kappa_2)$, $\bar{q}_{\text{snr}}(\kappa_1) \leq \bar{q}_{\text{snr}}(\kappa_2)$.*

A higher signal-to-noise ratio, therefore, shifts both extremal optimal rules weakly later in information time. Waiting for more information is cheaper in calendar time, so both extremal persuasion probabilities weakly rise.

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A Proofs for the Information-Clock Results

A.1 Proof of **Proposition 2.2**

Proof of Proposition 2.2. We first verify the Brownian-motion requirement. Under $\mathbb{P}^\ell$, the process $B_t^\ell = (X_t - \mu_\ell t)/\sigma$ is Brownian in $\mathbb{F}^X$. The product construction makes $\mathbb{U}$ independent of the entire evidence process. Hence, for $0 \leq s < t$, the increment $B_t^\ell - B_s^\ell$ is independent of $\mathcal{F}_s^{r,0} = \mathcal{F}_s^X \vee \mathcal{U}_s$ and is Gaussian with mean zero and variance $t - s$. Thus, B^ℓ is Brownian in $\mathbb{F}^{r,0}$. After the usual augmentation, B^ℓ remains a continuous martingale with quadratic variation t , so Lévy’s characterization implies that it is an $\mathbb{F}^r$ -Brownian motion. Since $\mathbb{F}^X \subseteq \tilde{\mathbb{F}} \subseteq \mathbb{F}^r$, it is also a $\tilde{\mathbb{F}}$ -Brownian motion.

We next verify the likelihood-ratio identity. Let $C \in \mathcal{F}_t^X$ and $D \in \mathcal{U}_t$. The product structure implies $\mathbb{P}^h(C \cap D) = \mathbb{P}_X^h(C)Q(D) = \mathbb{E}^\ell[Z_t \mathbb{1}_C]Q(D) = \mathbb{E}^\ell[Z_t \mathbb{1}_{C \cap D}]$. The events $C \cap D$ form a π -system that generates $\mathcal{F}_t^{r,0}$. The monotone-class theorem therefore implies $\frac{d\mathbb{P}^h}{d\mathbb{P}^\ell}\Big|_{\mathcal{F}_t^{r,0}} = Z_t$. If $A \in \bigcap_{u>t} \mathcal{F}_u^{r,0}$, then $\mathbb{P}^h(A) = \mathbb{E}^\ell[Z_u \mathbb{1}_A]$ for every $u > t$. Since $Z_u \rightarrow Z_t$ in $L^1(\mathbb{P}^\ell)$ as $u \downarrow t$, letting $u \downarrow t$ yields $\mathbb{P}^h(A) = \mathbb{E}^\ell[Z_t \mathbb{1}_A]$.

Because $\mathbb{P} = p_0\mathbb{P}^h + (1 - p_0)\mathbb{P}^\ell$ and both prior probabilities are positive, every $\mathbb{P}$ -null event is null under both state-contingent laws. The identity therefore extends to the completion, so $\frac{d\mathbb{P}^h}{d\mathbb{P}^\ell}\Big|_{\mathcal{G}_t} = Z_t$.

Finally, Z_t is $\mathcal{F}_t^X$ -measurable and therefore $\widetilde{\mathcal{F}}_t$ -measurable. Restricting the preceding identity to $\widetilde{\mathcal{F}}_t$ yields $\frac{d\mathbb{P}^h}{d\mathbb{P}^\ell}\Big|_{\widetilde{\mathcal{F}}_t} = Z_t$. This proves both parts of [Assumption 2.1](#). $\blacksquare$

A.2 Proof of [Proposition 3.2](#)

Proof of [Proposition 3.2](#). We proceed through two claims.

Claim A.1. *Every public stopping rule satisfies the stated lower bound on expected calendar time.*

Let $\mathbb{P}_{\tau, \mathbb{G}}^i$ denote the restriction of $\mathbb{P}^i$ to $\mathcal{G}_\tau$. Via Bayes' rule, $\log \frac{d\mathbb{P}_{\tau, \mathbb{G}}^h}{d\mathbb{P}_{\tau, \mathbb{G}}^\ell} = \log \frac{Q}{1-Q} - \log \frac{p_0}{1-p_0}$. Consequently, using $\mathbb{E}[Q] = p_0$,

$$\begin{aligned} & p_0 D(\mathbb{P}_{\tau, \mathbb{G}}^h \| \mathbb{P}_{\tau, \mathbb{G}}^\ell) + (1 - p_0) D(\mathbb{P}_{\tau, \mathbb{G}}^\ell \| \mathbb{P}_{\tau, \mathbb{G}}^h) \\ &= \mathbb{E} \left[(2Q - 1) \left(\log \frac{Q}{1-Q} - \log \frac{p_0}{1-p_0} \right) \right] = \mathbb{E}[\Psi(Q)] - \Psi(p_0). \end{aligned}$$

Let $\mathbb{P}_{\tau, \mathbb{F}}^i$ denote the restriction of $\mathbb{P}^i$ to $\mathcal{F}_\tau$. The stopped full likelihood ratio is Z_τ . To verify this claim, fix $A \in \mathcal{F}_\tau$. For every n , $A \cap \{\tau \leq n\} \in \mathcal{F}_n$, and bounded-time optional sampling delivers $\mathbb{P}^h(A \cap \{\tau \leq n\}) = \mathbb{E}^\ell[Z_n \mathbb{1}_{A \cap \{\tau \leq n\}}] = \mathbb{E}^\ell[Z_\tau \mathbb{1}_{A \cap \{\tau \leq n\}}]$. Letting $n \rightarrow \infty$ yields $\frac{d\mathbb{P}^h}{d\mathbb{P}^\ell}\Big|_{\mathcal{F}_\tau} = Z_\tau$. Under $\mathbb{P}^h$, $\log Z_\tau = \kappa B_\tau^h + \frac{\kappa^2}{2} \tau$. Since $\mathbb{E}^h[\tau] < \infty$, the stopped Brownian motion has zero mean, and therefore $D(\mathbb{P}_{\tau, \mathbb{F}}^h \| \mathbb{P}_{\tau, \mathbb{F}}^\ell) = \frac{\kappa^2}{2} \mathbb{E}^h[\tau]$. Similarly, $D(\mathbb{P}_{\tau, \mathbb{F}}^\ell \| \mathbb{P}_{\tau, \mathbb{F}}^h) = \frac{\kappa^2}{2} \mathbb{E}^\ell[\tau]$. The data-processing inequality for relative entropy and $\mathcal{G}_\tau \subseteq \mathcal{F}_\tau$ give

$$\mathbb{E}[\Psi(Q)] - \Psi(p_0) \leq \frac{\kappa^2}{2} \left(p_0 \mathbb{E}^h[\tau] + (1 - p_0) \mathbb{E}^\ell[\tau] \right) = \frac{\kappa^2}{2} \mathbb{E}[\tau].$$

Claim A.2. *For the full-speed posterior, expected stopping time is determined by the terminal posterior distribution through the stated identity.*

For the full-speed posterior M , direct differentiation produces $\Psi''(p) = \frac{1}{p^2(1-p)^2}$. Applying Itô's formula after localization yields

$$d\Psi(M_t) = \frac{\kappa^2}{2} dt + \kappa M_t(1 - M_t)\Psi'(M_t) dW_t.$$

The function $p \mapsto p(1-p)\Psi'(p) = 2p(1-p) \log \frac{p}{1-p} + 2p - 1$ is bounded on $(0, 1)$. Since $\mathbb{E}[S] < \infty$, the stopped stochastic integral is square integrable and has mean zero, and so passing through the localization, $\mathbb{E}[\Psi(M_S)] - \Psi(p_0) = \frac{\kappa^2}{2} \mathbb{E}[S]$. ■

A.3 Root's Timing Comparison

Lemma A.3. *Suppose η is locally Lipschitz, strictly positive in the interior of $\mathcal{I}$, and satisfies $\eta(x)^2 \leq C(1+x^2)$. Suppose the endpoints are either absorbing or inaccessible and that every finite inaccessible endpoint is approached with positive probability as time tends to infinity. Let μ_0 and μ_1 be centered integrable probability measures on $\overline{\mathcal{I}}$ satisfying $\mu_0 \preceq_{\text{cx}} \mu_1$. Let R be the Root stopping time that embeds μ_1 from μ_0 . Then $R \preceq_{\text{icx}} \rho$ for every uniformly integrable embedding ρ of μ_1 from μ_0 satisfying $\mathbb{E}[\rho] < \infty$.*

Proof of Lemma A.3. If $\mu_0 = \mu_1$, take $R = 0$. Since every embedding time is non-negative, $R \preceq_{\text{icx}} \rho$, so the conclusion follows. Hence, suppose $\mu_0 \neq \mu_1$.

Cox et al. (2019, Theorem 3.3), specialized to one marginal, states that the Root embedding minimizes $\mathbb{E}[\int_0^\rho f(s) ds]$ over all uniformly integrable embeddings whenever f is nonnegative and nondecreasing.

Fix $r \geq 0$. Let $(f_{r,n})_{n \geq 1}$ be nonnegative continuous nondecreasing functions satisfying $f_{r,n}(s) \uparrow \mathbb{1}_{\{s > r\}}$ for every $s \neq r$. Cox et al. (2019, Theorem 3.3) produces $\mathbb{E}[\int_0^R f_{r,n}(s) ds] \leq \mathbb{E}[\int_0^\rho f_{r,n}(s) ds]$. The monotone convergence theorem yields $\mathbb{E}[(R-r)^+] \leq \mathbb{E}[(\rho-r)^+]$. Since this holds for every $r \geq 0$, $R \preceq_{\text{icx}} \rho$. ■

A.4 Continuous Disclosure and the Canonical Clock

Suppose the public likelihood Λ is continuous and strictly positive. Define $N_t := \int_0^t \Lambda_s^{-1} d\Lambda_s$ and $I_t := \langle N \rangle_t / \kappa^2$. Under full transparency, $I_t = t$.

Lemma A.4 (Canonical clock for continuous disclosure). *Maintain [Assumption 2.1](#). If the public likelihood is continuous and strictly positive, then I is a disclosure clock in the sense of [Definition 3.6](#). In particular, after adding independent auxiliary randomness if necessary, there exist a usual filtration $\mathbb{H}$, an $\mathbb{H}$ -Brownian motion W , and an $[0, 1]$ -valued $\mathbb{H}$ -adapted process Q satisfying $dQ_u = \kappa Q_u(1 - Q_u) dW_u$ with $Q_0 = p_0$, such that each I_t is a finite $\mathbb{H}$ -stopping time, $\mathbb{G}_t \subseteq \mathcal{H}_{I_t}$, and $\hat{P}_t = Q_{I_t}$ for every $t \geq 0$. For every public stopping rule τ , the random time I_τ is an $\mathbb{H}$ -stopping time, $Q_{I_\tau} = \hat{P}_\tau$, and $(Q_{u \wedge I_\tau})_{u \geq 0}$ is uniformly integrable.*

Proof of [Lemma A.4](#). We proceed through four claims.

Claim A.5. *The normalized martingale part of the public posterior has quadratic variation I .*

Write $f(x) := p_0 x / (1 - p_0 + p_0 x)$, so $\hat{P}_t = f(\Lambda_t)$. Since $x f'(x) = f(x)(1 - f(x))$, Itô's formula for quadratic variation yields

$$d\langle \hat{P} \rangle_t = \hat{P}_t^2 (1 - \hat{P}_t)^2 d\langle N \rangle_t = \kappa^2 \hat{P}_t^2 (1 - \hat{P}_t)^2 dI_t.$$

Under $\mathbb{P}$, the process $\hat{P}$ is a bounded continuous $\mathbb{G}$ -martingale. Define

$$M_t := \int_0^t \frac{1}{\kappa \hat{P}_s (1 - \hat{P}_s)} d\hat{P}_s.$$

The public posterior remains in $(0, 1)$ at every finite time because Λ is strictly positive and finite. The integral is well defined by localization, and by the preceding quadratic-variation identity, $\langle M \rangle_t = I_t$.

Claim A.6. *On an enlarged probability space, there is a Brownian motion W such that $M_t = W_{I_t}$ for every calendar date t .*

Set $\zeta := I_\infty$ and, for $u \geq 0$, define $T_u := \inf\{t \geq 0: I_t > u\}$. Each T_u is a $\mathbb{G}$ -stopping time. On $\{\zeta < \infty\}$, the continuous local-martingale convergence theorem delivers a finite limit $M_\infty := \lim_{t \rightarrow \infty} M_t$. Define $\beta_u := M_{T_u}$, with the convention $M_{T_u} = M_\infty$ when $T_u = \infty$, and let $\mathbb{K} = (\mathcal{K}_u)_{u \geq 0}$ be the usual augmentation of the time-changed filtration $\mathcal{K}_u := \mathcal{G}_{T_u}$. The continuous local-martingale time-change theorem implies that β is a continuous $\mathbb{K}$ -local martingale, that ζ is a $\mathbb{K}$ -stopping time, and that $\langle \beta \rangle_u = u \wedge \zeta$.

Pass to a product extension carrying a Brownian motion $\widetilde{W}$ independent of $\mathcal{K}_\infty$, and let $\mathbb{H} = (\mathcal{H}_u)_{u \geq 0}$ be the usual augmentation of $(\mathcal{K}_u \vee \sigma(\widetilde{W}_s: 0 \leq s \leq u))_{u \geq 0}$. Define $W_u := \beta_u + \int_0^u \mathbb{1}_{\{s > \zeta\}} d\widetilde{W}_s$. The integrand is predictable because ζ is a $\mathbb{K}$ -stopping time. The process W is a continuous $\mathbb{H}$ -local martingale. Independence implies zero cross-variation between its two terms, so

$$\langle W \rangle_u = u \wedge \zeta + \int_0^u \mathbb{1}_{\{s > \zeta\}} ds = u.$$

Lévy's characterization theorem therefore implies that W is an $\mathbb{H}$ -Brownian motion. By construction, $\mathcal{G}_{T_u} \subseteq \mathcal{H}_u$ for every $u \geq 0$.

For every $t \geq 0$, $T_{I_t} \geq t$, and I is constant on $[t, T_{I_t}]$. Since a continuous local martingale is constant on every interval on which its quadratic variation is constant, $\beta_{I_t} = M_{T_{I_t}} = M_t$. The independent continuation does not move before ζ , and therefore $M_t = W_{I_t}$ for every $t \geq 0$.

Claim A.7. *Each I_t is a stopping time in the information-time filtration, and $\widehat{P}_t = Q_{I_t}$ for a full-speed posterior Q .*

For every $t \geq 0$, the identity $\{I_t \leq u\} = \{t \leq T_u\} \in \mathcal{G}_{T_u} \subseteq \mathcal{H}_u$ shows that I_t is an $\mathbb{H}$ -stopping time. If $A \in \mathcal{G}_t$, then $A \cap \{I_t \leq u\} = A \cap \{t \leq T_u\} \in \mathcal{G}_{T_u} \subseteq \mathcal{H}_u$, so $A \in \mathcal{H}_{I_t}$. Thus, $\mathcal{G}_t \subseteq \mathcal{H}_{I_t}$. Let Q be the unique $[0, 1]$ -valued strong solution to

$$dQ_u = \kappa Q_u(1 - Q_u) dW_u, \quad \text{with } Q_0 = p_0.$$

The stochastic time-change theorem implies $Q_{I_t} = p_0 + \int_0^t \kappa Q_{I_s}(1 - Q_{I_s}) dM_s$, while

$\hat{P}_t = p_0 + \int_0^t \kappa \hat{P}_s (1 - \hat{P}_s) dM_s$ by the definition of M . The coefficient is Lipschitz on $[0, 1]$, so pathwise uniqueness yields $\hat{P}_t = Q_{I_t}$ for every $t \geq 0$.

Claim A.8. *For every public stopping rule τ , the information time $U := I_\tau$ is a stopping time, $Q_U = \hat{P}_\tau$, and the stopped posterior is uniformly integrable.*

Continuity and monotonicity of I imply $I_s \leq u$ if and only if $s \leq T_u$. Consequently, for $U := I_\tau$, $\{U \leq u\} = \{\tau \leq T_u\} \in \mathcal{G}_{T_u} \subseteq \mathcal{H}_u$. Thus U is an $\mathbb{H}$ -stopping time. The process identity produces $Q_U = \hat{P}_\tau$. Finally, $Q \in [0, 1]$, so $(Q_{u \wedge U})_{u \geq 0}$ is bounded and therefore uniformly integrable. $\blacksquare$

A.5 The Mean of the Canonical Clock

Lemma A.9. *Maintain [Assumption 2.1](#). Let the public experiment have a continuous strictly positive likelihood, and let τ be a public stopping rule satisfying $\mathbb{E}[\tau] < \infty$. Then I_τ is integrable and $\frac{\kappa^2}{2} \mathbb{E}[I_\tau] = \mathbb{E}[\Psi(\hat{P}_\tau)] - \Psi(p_0) \leq \frac{\kappa^2}{2} \mathbb{E}[\tau]$.*

Proof of [Lemma A.9](#). Apply [Lemma A.4](#), and retain its notation Q . For every $n \geq 1$, define

$$\xi_n := \inf\{t \geq 0: I_t \geq n\}, \quad \tau_n := \tau \wedge n \wedge \xi_n, \quad \text{and} \quad U_n := I_{\tau_n}.$$

The random time τ_n is a bounded public stopping rule, and $0 \leq U_n \leq n$. By [Lemma A.4](#), U_n is a bounded stopping time for Q and $Q_{U_n} = \hat{P}_{\tau_n}$. The full-speed identity in [Proposition 3.2](#) implies $\mathbb{E}[U_n] = \frac{2}{\kappa^2} (\mathbb{E}[\Psi(\hat{P}_{\tau_n})] - \Psi(p_0))$, while the public mean-time lower bound implies $\mathbb{E}[U_n] \leq \mathbb{E}[\tau_n]$. As $n \rightarrow \infty$, $\tau_n \uparrow \tau$ and $U_n \uparrow I_\tau$. Hence, the monotone convergence theorem yields $\mathbb{E}[I_\tau] \leq \mathbb{E}[\tau] < \infty$. Applying the full-speed identity in [Proposition 3.2](#) to (Q, I_τ) , and using $Q_{I_\tau} = \hat{P}_\tau$, we have $\frac{\kappa^2}{2} \mathbb{E}[I_\tau] = \mathbb{E}[\Psi(\hat{P}_\tau)] - \Psi(p_0)$, proving the result. $\blacksquare$

A.6 Proof of Theorem 3.7

Proof of Theorem 3.7. Fix the disclosure clock C in the theorem, retain the corresponding filtration $\mathbb{J}$ and full-speed posterior Q , and set $U := C_\tau$. By Definition 3.6, U is a stopping time for Q and $Q_U = \hat{P}_\tau$. Clock domination gives $U \leq \tau$, so U is integrable. Because $Q \in [0, 1]$, $(Q_{u \wedge U})_{u \geq 0}$ is uniformly integrable. We proceed through three claims.

Claim A.10. *The shifted full-speed posterior satisfies the hypotheses of Lemma A.3.*

Shift the posterior by setting $\tilde{Q}_u := Q_u - p_0$. Its state interval is $\mathcal{I} = (-p_0, 1 - p_0)$, and it satisfies

$$d\tilde{Q}_u = \eta(\tilde{Q}_u) dW_u, \quad \text{for } \eta(x) := \kappa(x + p_0)(1 - p_0 - x).$$

The coefficient η is locally Lipschitz, strictly positive in the interior of $\mathcal{I}$, and bounded. The endpoints are inaccessible. Indeed, by Itô's formula,

$$d \left(\log \frac{Q_u}{1 - Q_u} \right) = \kappa dW_u + \frac{\kappa^2}{2} (2Q_u - 1) du.$$

The drift is bounded and the volatility is constant, so the log odds cannot explode at a finite time.

The processes P and Q solve the same stochastic differential equation from p_0 , so they have the same law. Under $\mathbb{P}^\ell$,

$$\log \frac{P_u}{1 - P_u} = \log \frac{p_0}{1 - p_0} + \kappa B_u^\ell - \frac{\kappa^2}{2} u.$$

The strong law of large numbers for Brownian motion implies $B_u^\ell/u \rightarrow 0$ almost surely, so $P_u \rightarrow 0$ under $\mathbb{P}^\ell$. Under $\mathbb{P}^h$, $B_u^\ell = B_u^h + \kappa u$, and hence $P_u \rightarrow 1$. Therefore, $\tilde{Q}$ approaches each finite inaccessible endpoint with positive probability as $u \rightarrow \infty$.

Claim A.11. *Root stopping produces ν in the full-speed posterior Q at a time $R_Q \preceq_{\text{cx}} U$.*

Let $\mu_0 := \delta_0$ and $\mu_1 := \mathcal{L}(\tilde{Q}_U)$. Uniform integrability gives $\int x \mu_1(dx) = \mathbb{E}[\tilde{Q}_U] = 0$, so Jensen's inequality implies $\mu_0 \preceq_{\text{cx}} \mu_1$. Applying [Lemma A.3](#) to $\tilde{Q}$, and translating the resulting barrier by p_0 , gives a deterministic Root barrier $\mathcal{B}^\nu$ for Q . Set $R_Q := R_{\mathcal{B}^\nu}(Q)$. Then $R_Q \preceq_{\text{icx}} U$ and $\mathcal{L}(Q_{R_Q}) = \nu$. Since U is integrable, so is R_Q . Applying the full-speed identity in [Proposition 3.2](#) to (Q, R_Q) and (Q, U) , and using equality of their terminal laws, gives $\mathbb{E}[R_Q] = \mathbb{E}[U]$. Hence, $R_Q \preceq_{\text{cx}} U$.

Claim A.12. *The same deterministic barrier produces a transparent stopping time R^ν with the required terminal distribution and timing law.*

Use the same barrier for the fully informed posterior and set $R^\nu := R_{\mathcal{B}^\nu}(P)$. Since P and Q solve the same stochastic differential equation from the same initial condition, $\mathcal{L}(R^\nu, P_{R^\nu}) = \mathcal{L}(R_Q, Q_{R_Q})$. Thus, $\mathcal{L}(P_{R^\nu}) = \nu$ and $R^\nu \preceq_{\text{cx}} C_\tau$. Clock domination gives $C_\tau \leq \tau$ almost surely, and therefore $R^\nu \preceq_{\text{icx}} \tau$.

If $\mathbb{E}[c(\tau)] = \infty$, boundedness of F makes the original payoff $-\infty$, so the payoff inequality is immediate. Otherwise, $\mathbb{E}[c(R^\nu)] \leq \mathbb{E}[c(C_\tau)] \leq \mathbb{E}[c(\tau)]$, while equality of the terminal posterior distributions gives $\mathbb{E}[F(P_{R^\nu})] = \mathbb{E}[F(\hat{P}_\tau)]$.

Full transparency is clock-dominated with disclosure clock $C_t = t$, so $V_{F,c}^{\text{garb}} \geq V_{F,c}^0$. Conversely, the transparent replacement of every policy in the domain of $V_{F,c}^{\text{garb}}$ lies in the domain of $V_{F,c}^0$ and yields weakly higher expected payoff. Taking suprema gives $V_{F,c}^{\text{garb}} \leq V_{F,c}^0$.

If $\tau \leq T$, then $C_\tau \leq T$. Convex order gives $\mathbb{E}[(R^\nu - T)^+] \leq \mathbb{E}[(C_\tau - T)^+] = 0$, and hence $R^\nu \leq T$ almost surely.

If c is strictly increasing and $\mathbb{P}(C_\tau < \tau) > 0$, then $c(C_\tau) \leq c(\tau)$ almost surely, with strict inequality on a set of positive probability. Therefore, $\mathbb{E}[c(R^\nu)] \leq \mathbb{E}[c(C_\tau)] < \mathbb{E}[c(\tau)]$. Finally, if c is strictly convex and $\mathcal{L}(R^\nu) \neq \mathcal{L}(C_\tau)$, strict Jensen's inequality under a martingale coupling for $R^\nu \preceq_{\text{cx}} C_\tau$ gives $\mathbb{E}[c(R^\nu)] < \mathbb{E}[c(C_\tau)]$. $\blacksquare$

A.7 Stored Disclosure and Discontinuous Clocks

We now give a delayed-disclosure construction in which the underlying posterior is continuous in evidence time, while its release in calendar time may be delayed or batched.

Definition A.13 (Stored-disclosure experiment). A stored-disclosure experiment consists of a usual filtration $\mathbb{K} = (\mathcal{K}_u)_{u \geq 0}$ satisfying $\mathcal{K}_u \subseteq \mathcal{F}_u$, together with an increasing right-continuous family $A = (A_t)_{t \geq 0}$, with $A_0 = 0$. The evidence-time posterior $\check{P}_u := \mathbb{P}(\mu = \mu_h \mid \mathcal{K}_u)$ is required to have continuous paths in $(0, 1)$. Each A_t is a finite $\mathbb{K}$ -stopping time satisfying $A_t \leq t$. The public filtration $\mathbb{G}^A = (\mathcal{G}_t^A)_{t \geq 0}$ is the usual augmentation of

$$\mathcal{G}_t^{A,0} := \sigma(A_s : 0 \leq s \leq t) \vee \mathcal{K}_{A_t},$$

and its posterior is $\hat{P}_t^A := \mathbb{P}(\mu = \mu_h \mid \mathcal{G}_t^A)$. Define

$$\Gamma_u := \int_0^u \frac{d\langle \check{P} \rangle_s}{\kappa^2 \check{P}_s^2 (1 - \check{P}_s)^2}, \quad C_t := \Gamma_{A_t}.$$

The process A_t is the latest date of the underlying record released by calendar date t . A jump from a to b releases the record generated over $(a, b]$, rather than a summary constructed at the release date. The disclosure clock $C_t = \Gamma_{A_t}$ may therefore jump even though Γ is continuous.

Proposition A.14 (Stored disclosure admits a disclosure clock). *Maintain **Assumption 2.1**. Every stored-disclosure experiment satisfies $\hat{P}_t^A = \check{P}_{A_t}$ for every $t \geq 0$. Moreover, on an independent extension there exist a usual filtration $\mathbb{J}$, a full-speed posterior Q , and the disclosure clock $C_t = \Gamma_{A_t}$ such that $\hat{P}_t^A = Q_{C_t}$, $\mathcal{G}_t^A \subseteq \mathcal{J}_{C_t}$, each C_t is a $\mathbb{J}$ -stopping time, and C_τ is a $\mathbb{J}$ -stopping time for every public stopping rule τ . Consequently, if $C_t \leq t$ for every t almost surely, the experiment is clock-dominated in the sense of **Definition 3.6**.*

Proof. Because $\check{P}_u = \mathbb{E}[\mathbb{1}_{\{\mu=\mu_h\}} \mid \mathcal{K}_u]$ and A_t is a finite $\mathbb{K}$ -stopping time, optional sampling at the stopped sigma-field gives $\mathbb{E}[\mathbb{1}_{\{\mu=\mu_h\}} \mid \mathcal{K}_{A_t}] = \check{P}_{A_t}$. If $s \leq t$, then $A_s \leq A_t$, so $\mathcal{K}_{A_s} \subseteq \mathcal{K}_{A_t}$. Since A_s is $\mathcal{K}_{A_s}$ -measurable, the release history through date t belongs to $\mathcal{K}_{A_t}$. Right-continuity and completion preserve this inclusion, and hence $\mathcal{G}_t^A \subseteq \mathcal{K}_{A_t}$. The posterior $\check{P}_{A_t}$ is public at date t , so the tower property gives $\hat{P}_t^A = \check{P}_{A_t}$.

Define $M_u := \int_0^u \frac{1}{\kappa \check{P}_s(1-\check{P}_s)} d\check{P}_s$. Then M is a continuous $\mathbb{K}$ -local martingale satisfying $\langle M \rangle_u = \Gamma_u$. For $v \geq 0$, set $D_v := \inf\{u \geq 0 : \Gamma_u > v\}$. The continuous local-martingale time-change theorem, followed by an independent Brownian continuation beyond Γ_∞ , produces a usual filtration $\mathbb{J}$, a $\mathbb{J}$ -Brownian motion, and a full-speed posterior Q such that $\check{P}_u = Q_{\Gamma_u}$ and $\mathcal{K}_{D_v} \subseteq \mathcal{J}_v$.

Continuity and monotonicity of Γ give $\{C_t \leq v\} = \{A_t \leq D_v\}$. Since A_t and D_v are $\mathbb{K}$ -stopping times, the event on the right belongs to $\mathcal{K}_{D_v} \subseteq \mathcal{J}_v$, so each C_t is a $\mathbb{J}$ -stopping time. If $B \in \mathcal{G}_t^A$, then, for every $v \geq 0$,

$$B \cap \{C_t \leq v\} = B \cap \{A_t \leq D_v\} \in \mathcal{K}_{D_v} \subseteq \mathcal{J}_v.$$

Thus, $\mathcal{G}_t^A \subseteq \mathcal{J}_{C_t}$, and $\hat{P}_t^A = \check{P}_{A_t} = Q_{C_t}$.

Finally, if τ is a public stopping rule, right-continuity and monotonicity of C give, for every $v > 0$,

$$\{C_\tau < v\} = \bigcup_{r \in \mathbb{Q}_+} (\{\tau < r\} \cap \{C_r < v\}).$$

The inclusion $\mathcal{G}_r^A \subseteq \mathcal{J}_{C_r}$ makes every event in the union $\mathcal{J}_v$ -measurable. Right-continuity of $\mathbb{J}$ therefore makes C_τ a $\mathbb{J}$ -stopping time. $\blacksquare$

B Proofs for the Receiver-Stopping Results

B.1 Proof of Lemma 4.3

Proof of Lemma 4.3. We proceed through three claims.

Claim B.1. *For every $u \geq 0$, $C_{\theta_u} = u \wedge U$, $\hat{P}_{\theta_u}^S = Q_{u \wedge U}$, and U is an $\mathbb{H}$ -stopping time.*

Continuity and monotonicity of C imply that T_u^+ is a $\mathbb{G}^S$ -stopping time. On $\{U \leq u\}$, the clock does not move strictly above u before ρ , so $\theta_u = \rho$ and $C_{\theta_u} = U$. On $\{U > u\}$, continuity gives $T_u^+ < \rho$ and $C_{T_u^+} = u$. Hence, $C_{\theta_u} = u \wedge U$. The identity in Definition 3.6 then yields $\hat{P}_{\theta_u}^S = Q_{u \wedge U}$.

Moreover, $\{U \leq u\} = \{\rho \leq T_u^+\}$. For two stopping times, the event that one precedes the other belongs to the sigma-field at their minimum. Therefore, $\{U \leq u\} \in \mathcal{G}_{\theta_u}^S = \mathcal{H}_u^0 \subseteq \mathcal{H}_u$, proving that U is an $\mathbb{H}$ -stopping time.

Claim B.2. *If $V \leq U$ is an $\mathbb{H}$ -stopping time, then θ_V is a $\mathbb{G}^S$ -stopping time.*

Let $V \leq U$ be an $\mathbb{H}$ -stopping time. For $n \geq 1$, define the upper dyadic approximation $V_n := 2^{-n}(\lfloor 2^n V \rfloor + 1)$. Then $V_n \downarrow V$. Right-continuity of $\mathbb{H}$ makes every V_n an $\mathbb{H}$ -stopping time. For each n , the random time V_n has countable range. Since $\mathbb{H}$ is the completion of $\mathbb{H}^0$, after modifying on one null set we may take each event $\{V_n = u\}$ in $\mathcal{H}_u^0 = \mathcal{G}_{\theta_u}^S$. Pasting the stopping times θ_u on these events shows that θ_{V_n} is a $\mathbb{G}^S$ -stopping time. The generalized inverse $u \mapsto T_u^+$ is right-continuous because C is continuous and nondecreasing. Thus, $\theta_{V_n} \downarrow \theta_V$. Right-continuity of $\mathbb{G}^S$ implies that θ_V is a stopping time.

Claim B.3. *The identities $\hat{P}_{\theta_V}^S = Q_V$ and $\theta_V - V \leq \rho - U$ hold almost surely.*

On the full-probability event on which C is continuous and unbanked, the identity $C_{\theta_u} = u \wedge U$ holds for every $u \geq 0$. Evaluating it at V gives $C_{\theta_V} = V$ and $\hat{P}_{\theta_V}^S = Q_V$.

On $\{V < U\}$, $\theta_V < \rho$. Unbankedness applied between θ_V and ρ gives $U - V = C_\rho - C_{\theta_V} \leq \rho - \theta_V$. Rearranging yields $\theta_V - V \leq \rho - U$. On $\{V = U\}$, the definition gives $\theta_V = \rho$, so equality holds. Since $V \leq U$, the same inequality also gives $\theta_V \leq \rho$; in particular, θ_V is integrable. $\blacksquare$

B.2 Proof of Lemma 4.4

Proof of Lemma 4.4. We proceed through four claims.

Claim B.4. *For every $t \geq 0$ and $B \in \mathcal{F}_\infty^Q$, $\mathbb{E}[\mathbb{1}_B \mid \mathcal{G}_t] = \mathbb{E}[\mathbb{1}_B \mid \mathcal{F}_t^Q]$.*

Retain $\mathbb{J}$ and Q fixed in §4. The process Q is driven by a $\mathbb{J}$ -Brownian motion W . Since $Q_u \in (0, 1)$ at every finite date, $W_t = \int_0^t \frac{1}{\kappa Q_s(1-Q_s)} dQ_s$, so the path of Q determines the path of W . Thus, $\mathbb{F}^Q$ is the usual natural filtration of W .

Fix $B \in \mathcal{F}_\infty^Q$, and set $M_t := \mathbb{E}[\mathbb{1}_B \mid \mathcal{F}_t^Q]$. By the martingale representation theorem for Brownian motion, there is an $\mathbb{F}^Q$ -predictable process h such that $M_t = M_0 + \int_0^t h_s dW_s$. Since $\mathbb{F}^Q \subseteq \mathbb{J}$, the process h is also $\mathbb{J}$ -predictable. Because W is a $\mathbb{J}$ -Brownian motion, M is a $\mathbb{J}$ -martingale. Moreover, $M_t \rightarrow \mathbb{1}_B$ in L^1 . Hence, $M_t = \mathbb{E}[\mathbb{1}_B \mid \mathcal{G}_t]$, proving the claim.

Claim B.5. *There is a probability kernel K such that $K(Q, \cdot)$ is a conditional distribution of U given $\mathcal{F}_\infty^Q$ and $K(Q, [0, t])$ is $\mathcal{F}_t^Q$ -measurable for every $t \geq 0$.*

Continuous path space, equipped with uniform convergence on compact intervals, is a complete separable metric space. A regular conditional distribution of U given $\mathcal{F}_\infty^Q$ therefore exists. This is a measurable map that assigns a probability distribution over stopping dates to each posterior path. Write it as $K(Q, ds)$. For every $t \geq 0$, $K(Q, [0, t]) = \mathbb{E}[\mathbb{1}_{\{U \leq t\}} \mid \mathcal{F}_\infty^Q]$.

The event $\{U \leq t\}$ belongs to $\mathcal{G}_t$. Fix $B \in \mathcal{F}_\infty^Q$. The tower property gives $\mathbb{E}[\mathbb{1}_{\{U \leq t\}} \mathbb{1}_B] = \mathbb{E}[\mathbb{1}_{\{U \leq t\}} \mathbb{E}[\mathbb{1}_B \mid \mathcal{G}_t]]$. By Claim B.4, this equals $\mathbb{E}[\mathbb{1}_{\{U \leq t\}} \mathbb{E}[\mathbb{1}_B \mid \mathcal{F}_t^Q]]$, which in turn equals $\mathbb{E}[\mathbb{E}[\mathbb{1}_{\{U \leq t\}} \mid \mathcal{F}_t^Q] \mathbb{1}_B]$. Hence, $\mathbb{E}[\mathbb{1}_{\{U \leq t\}} \mid \mathcal{F}_\infty^Q] = \mathbb{E}[\mathbb{1}_{\{U \leq t\}} \mid \mathcal{F}_t^Q]$, so $K(Q, [0, t])$ is $\mathcal{F}_t^Q$ -measurable. Choose the kernel so that this measurability holds simultaneously at every rational date, and use right-continuity of its distribution function to extend it to every date.

Claim B.6. *Applying the same kernel to P produces σ_D such that $((P_u)_{u \geq 0}, \sigma_D) \stackrel{d}{=} ((Q_u)_{u \geq 0}, U)$.*

By uniqueness in law for (2), P and Q have the same distribution on continuous path space. Let ξ_D be uniform on $[0, 1]$ and independent of $\sigma(\mu, \mathcal{F}_\infty)$, and define $\sigma_D := \inf\{t \geq 0: K(P, [0, t]) \geq \xi_D\}$. Since $K(P, [0, t])$ is measurable from the path of P through t , σ_D is a stopping time in the filtration generated by P and ξ_D . Conditional on the path of P , its law is $K(P, \cdot)$. Therefore, $((P_u)_{u \geq 0}, \sigma_D) \stackrel{d}{=} ((Q_u)_{u \geq 0}, U)$. By the definitions of Z^D and Z^Q in Lemma 4.4, this equality also gives $Z^D \stackrel{d}{=} Z^Q$.

Claim B.7. *Every stopping time $\eta \leq \sigma_D$ for the usual natural filtration of Z^D has a stopping time $V \leq U$ for the usual natural filtration of Z^Q such that $(P_\eta, \eta) \stackrel{d}{=} (Q_V, V)$.*

Regard Z^D and Z^Q as random elements of the same path space. Let $\mathbb{E}^0 = (\mathcal{E}_t^0)_{t \geq 0}$ be the filtration generated directly by the coordinate paths, before completion or right-continuous augmentation, and let $\mathbb{E} = (\mathcal{E}_t)_{t \geq 0}$ be its right-continuous modification.

For each $q \in \mathbb{Q}_+$, there is $B_q \in \mathcal{E}_q^0$ such that $\{\eta < q\} = \{Z^D \in B_q\}$ almost surely. Indeed, $\{\eta < q\} = \bigcup_{r \in \mathbb{Q}_+, r < q} \{\eta \leq r\}$. For each $r < q$, choose $s \in (r, q)$. Because η is a stopping time for the usual augmentation of the natural filtration of Z^D , the event $\{\eta \leq r\}$ agrees almost surely with the inverse image of an event in $\mathcal{E}_s^0 \subseteq \mathcal{E}_q^0$. Taking the countable union produces B_q .

Define $A_q := \bigcup_{r \in \mathbb{Q}_+, r < q} B_r$. Then $(A_q)_{q \in \mathbb{Q}_+}$ is increasing, $A_q \in \mathcal{E}_q^0$, and $\{Z^D \in A_q\} = \{\eta < q\}$ almost surely for every $q \in \mathbb{Q}_+$. Since $\mathbb{Q}_+$ is countable, these identities hold simultaneously outside one null set. Define $\varphi(z) := \inf\{q \in \mathbb{Q}_+: z \in A_q\}$, where $\inf \emptyset = \infty$. For every $t \geq 0$, $\{\varphi < t\} = \bigcup_{q \in \mathbb{Q}_+, q < t} A_q \in \mathcal{E}_t^0$. Right-continuity of $\mathbb{E}$ therefore makes φ an $\mathbb{E}$ -stopping time. For every $q \in \mathbb{Q}_+$, density of $\mathbb{Q}_+$ and the definition of A_q give $\{\varphi < q\} = A_q$. Hence, $\varphi(Z^D) = \eta$ almost surely.

Writing a coordinate path as $z = (x, d)$, define $\zeta(z) := \inf\{u \geq 0: d_u = 1\}$. The identity $\{\zeta \leq t\} = \{d_t = 1\}$ shows that ζ is an $\mathbb{E}^0$ -stopping time. Replace φ by $\varphi \wedge \zeta$. Since $\eta \leq \sigma_D = \zeta(Z^D)$ almost surely, this replacement preserves $\varphi(Z^D) = \eta$ almost surely and gives $\varphi \leq \zeta$ on the path space. Set $V := \varphi(Z^Q)$. For every $t \geq 0$, the event $\{V \leq t\}$ belongs to the usual natural filtration of Z^Q at t , so V is a stopping time

for that filtration. Moreover, $\varphi \leq \zeta$ gives $V \leq U$. Finally, $Z^D \stackrel{d}{=} Z^Q$ and $\varphi(Z^D) = \eta$ yield $(Z^D, \eta) \stackrel{d}{=} (Z^Q, V)$. Since the first-coordinate paths are continuous, evaluating them at the corresponding stopping times gives $(P_\eta, \eta) \stackrel{d}{=} (Q_V, V)$. $\blacksquare$

B.3 Proof of Theorem 4.6

Proof of Theorem 4.6. We proceed through three claims.

Claim B.8. *The transparent acceleration is clock-dominated and has an unbanked disclosure clock.*

Let $\tilde{\mathbb{F}}^P = (\tilde{\mathcal{F}}_u^P)_{u \geq 0}$ be the usual augmentation of the filtration generated by P and the independent date-zero randomization used to construct σ_D . Take $Q^D := P$ and $C_t^D := t \wedge \sigma_D$. Since the added randomization is independent of the state, $P_u = \mathbb{E}[\mathbb{1}_{\{\mu = \mu_h\}} \mid \tilde{\mathcal{F}}_u^P]$. Optional sampling gives $\mathbb{E}[\mathbb{1}_{\{\mu = \mu_h\}} \mid \tilde{\mathcal{F}}_{t \wedge \sigma_D}^P] = P_{t \wedge \sigma_D}$.

The raw public sigma-field $\mathcal{G}_t^{D,0}$ is contained in $\tilde{\mathcal{F}}_{t \wedge \sigma_D}^P$, and the inclusion survives completion and right-continuous augmentation. Moreover, $P_{t \wedge \sigma_D}$ is $\mathcal{G}_t^D$ -measurable by Definition 4.5. The tower property therefore gives

$$\hat{P}_t^D = \mathbb{E} \left[\mathbb{E} \left[\mathbb{1}_{\{\mu = \mu_h\}} \mid \tilde{\mathcal{F}}_{t \wedge \sigma_D}^P \right] \middle| \mathcal{G}_t^D \right] = P_{t \wedge \sigma_D} = Q_{C_t^D}^D.$$

Thus, $\mathcal{G}_t^D \subseteq \tilde{\mathcal{F}}_{C_t^D}^P$. Each C_t^D is a $\tilde{\mathbb{F}}^P$ -stopping time. Moreover, if τ is a $\mathbb{G}^D$ -stopping time, then $C_\tau^D = \tau \wedge \sigma_D$ is a $\tilde{\mathbb{F}}^P$ -stopping time. Hence, C^D is a disclosure clock. It satisfies, almost surely, $C_t^D - C_s^D \leq t - s$ for every $0 \leq s \leq t$, so it is unbanked and, in particular, clock-dominated.

Claim B.9. *The receiver optimally waits until σ_D under the transparent acceleration.*

By Lemma 4.4, $\mathbb{E}[\sigma_D] = \mathbb{E}[U] \leq \mathbb{E}[\rho] < \infty$. Fix an integrable $\mathbb{G}^D$ -stopping time η . Since the public posterior is constant after σ_D and $\kappa_R > 0$, replacing η by $\eta \wedge \sigma_D$ weakly raises the receiver's payoff. We may therefore suppose that $\eta \leq \sigma_D$.

By Lemma 4.4, there is a stopping time $V \leq U$ for the natural filtration of Z^Q such that $(P_\eta, \eta) \stackrel{d}{=} (Q_V, V)$. By Lemma 4.3, both coordinate processes of Z^Q are

$\mathbb{H}$ -adapted, so V is an $\mathbb{H}$ -stopping time. The same lemma makes θ_V a $\mathbb{G}^S$ -stopping time, with $\hat{P}_{\theta_V}^S = Q_V$ and $\theta_V - V \leq \rho - U$. Optimality of ρ gives $\mathbb{E}[r(Q_V) - \kappa_R \theta_V] \leq \mathbb{E}[r(Q_U) - \kappa_R \rho]$. Therefore,

$$\begin{aligned} \mathbb{E}[r(P_\eta) - \kappa_R \eta] &= \mathbb{E}[r(Q_V) - \kappa_R V] \\ &= \mathbb{E}[r(Q_V) - \kappa_R \theta_V] + \kappa_R \mathbb{E}[\theta_V - V] \\ &\leq \mathbb{E}[r(Q_U) - \kappa_R \rho] + \kappa_R \mathbb{E}[\rho - U] \\ &= \mathbb{E}[r(Q_U) - \kappa_R U] = \mathbb{E}[r(P_{\sigma_D}) - \kappa_R \sigma_D]. \end{aligned}$$

Thus, σ_D is a receiver best response to D .

Claim B.10. *The transparent acceleration weakly raises the sender's payoff and gives the receiver the stated gain.*

By Lemma 4.4, $\mathcal{L}(P_{\sigma_D}) = \mathcal{L}(Q_U) = \mathcal{L}(\hat{P}_\rho^S)$ and $\mathcal{L}(\sigma_D) = \mathcal{L}(U)$. Since $U = C_\rho \leq \rho$ and c is increasing, $\mathbb{E}[c(\sigma_D)] = \mathbb{E}[c(U)] \leq \mathbb{E}[c(\rho)]$. Equality of the terminal posterior distributions proves the sender's payoff comparison. Finally,

$$\mathbb{E}[r(P_{\sigma_D}) - \kappa_R \sigma_D] = \mathbb{E}[r(\hat{P}_\rho^S) - \kappa_R \rho] + \kappa_R \mathbb{E}[\rho - C_\rho],$$

which proves the receiver's gain. ■

B.4 Proof of Corollary 4.7

Proof of Corollary 4.7. The conclusion follows from Theorem 4.6. ■

C Proofs for the Comparative Statics

C.1 Proof of Lemma 5.1

Proof of Lemma 5.1. Continuity of P and $p_0 < \bar{P}$ give

$$\mathbb{1}_{\{P_\rho \geq \bar{P}\}} \leq \mathbb{1}_{\{H \leq \rho\}} = \mathbb{1}_{\{P_{\rho \wedge H} \geq \bar{P}\}}.$$

Moreover, $\rho \wedge H \leq \rho$, so monotonicity of c gives $c(\rho \wedge H) \leq c(\rho)$. Hence, $\rho \wedge H$ weakly dominates ρ .

Suppose ρ is transparent-optimal. The payoff gain from replacing ρ by $\rho \wedge H$ is the expectation of $\mathbb{1}_{\{H \leq \rho\}} - \mathbb{1}_{\{P_\rho \geq \bar{P}\}} + c(\rho) - c(\rho \wedge H)$, which is nonnegative almost surely. Optimality forces this expectation to equal zero. This random variable therefore vanishes almost surely, so the persuasion indicators coincide. Thus, $\rho \wedge H$ is optimal and preserves the persuasion probability. $\blacksquare$

C.2 Proof of Lemma 5.2

Proof of Lemma 5.2. We proceed through four claims.

Claim C.1. *For every finite horizon n , an optimal stopping rule exists; the finite-horizon optimizer set is closed under pairwise minimum and maximum and has a least element ρ_n .*

Define the adapted reward process $G_t := \mathbb{1}_{\{H \leq t\}} - c(t)$. For each integer $n \geq 1$, let $V_n := \sup_{\rho \leq n} \mathbb{E}[G_\rho]$, where the supremum is over $\mathbb{F}^P$ -stopping times. The sequence $(V_n)_{n \geq 1}$ is nondecreasing and bounded above by one. On $[0, n]$, the process G is bounded and càdlàg. If stopping times $\theta_k \uparrow \theta \leq n$, continuity of c and the fact that $t \mapsto \mathbb{1}_{\{H \leq t\}}$ has only an upward jump imply $\limsup_{k \rightarrow \infty} G_{\theta_k} \leq G_\theta$ almost surely. If $\theta_k \downarrow \theta$, right-continuity gives $G_{\theta_k} \rightarrow G_\theta$ almost surely. Boundedness turns both pathwise statements into the corresponding statements in expectation. The finite-horizon Snell-envelope theorem therefore yields an optimal stopping time. The set of horizon- n optimizers is closed under pairwise minimum and maximum. Indeed, if θ and η are horizon- n optimal, then $G_{\theta \wedge \eta} + G_{\theta \vee \eta} = G_\theta + G_\eta$ pathwise. Each expected payoff on the left is at most V_n , while their sum equals $2V_n$, so both stopping times are optimal. Because the optimizer set is closed under pairwise minima, there is a decreasing sequence of horizon- n optimizers whose limit is its essential infimum, namely the largest random time that is almost surely no greater than every horizon- n opti-

mizer. Right-continuity of $\mathbb{F}^P$ makes the limit a stopping time, and right-continuity and boundedness of G give optimality by the bounded convergence theorem. Let ρ_n denote this least horizon- n optimizer.

Claim C.2. *The least finite-horizon optimizers are increasing in the horizon, and their limit is an integrable global optimizer.*

The stopping time $\rho_n \wedge H$ weakly dominates ρ_n in the horizon- n problem and is therefore also optimal. Minimality of ρ_n gives $\rho_n \leq \rho_n \wedge H \leq \rho_n$, so $\rho_n \leq H$ almost surely.

The least finite-horizon optimizers are increasing in the horizon. Fix $m > n$, and set $a := \rho_n \wedge \rho_m$ and $b := \rho_n \vee \rho_m$. Then $a \leq n$, $b \leq m$, and the unordered pair $\{a, b\}$ equals $\{\rho_n, \rho_m\}$ pathwise. Hence, $\mathbb{E}[G_a] + \mathbb{E}[G_b] = V_n + V_m$. Feasibility gives $\mathbb{E}[G_a] \leq V_n$ and $\mathbb{E}[G_b] \leq V_m$, so both inequalities are equalities. Thus, a is horizon- n optimal. Since ρ_n is the least horizon- n optimizer, $\rho_n \leq a \leq \rho_n$, and therefore $\rho_n \leq \rho_m$ almost surely.

Set $\rho^* := \lim_{n \rightarrow \infty} \rho_n$. Immediate stopping yields zero, so $V_n \geq 0$. Since $\rho_n \leq H$, $0 \leq V_n = \mathbb{P}(\rho_n = H) - \mathbb{E}[c(\rho_n)]$, and therefore $\mathbb{E}[c(\rho_n)] \leq 1$. The monotone convergence theorem gives $\mathbb{E}[c(\rho^*)] \leq 1$. Because c is nonzero, choose $t_0 > 0$ with $c(t_0) > 0$. Convexity and $c(0) = 0$ imply $c(t) \geq \frac{c(t_0)}{t_0}t$ for every $t \geq t_0$, and hence $t \leq t_0 + \frac{t_0}{c(t_0)}c(t)$ for every $t \geq 0$. Thus, $\mathbb{E}[\rho^*] < \infty$, so $\rho^* \in \mathcal{T}_H$.

The events $\{\rho_n = H\}$ are increasing, and their union is contained in $\{\rho^* = H\}$. Hence, $J_c(\rho^*) \geq \lim_{n \rightarrow \infty} V_n$. To identify the limit, fix $\rho \in \mathcal{T}_H$. Then $\rho \wedge n \leq n$, $\mathbb{1}_{\{\rho \wedge n = H\}} \rightarrow \mathbb{1}_{\{\rho = H\}}$ almost surely, and $c(\rho \wedge n) \uparrow c(\rho)$. Therefore, $J_c(\rho \wedge n) \rightarrow J_c(\rho)$, including when the limit equals $-\infty$. Since $V_n \geq J_c(\rho \wedge n)$, taking limits and then the supremum over $\rho \in \mathcal{T}_H$ gives $\lim_n V_n \geq \mathcal{V}(c)$. The reverse inequality follows from $\rho_n \in \mathcal{T}_H$. Thus, $\lim_n V_n = \mathcal{V}(c)$, and feasibility of ρ^* forces $J_c(\rho^*) = \mathcal{V}(c)$. Therefore, $\mathcal{S}(c)$ is nonempty.

Claim C.3. *The global optimizer set $\mathcal{S}(c)$ is closed under pairwise minimum and*

maximum.

For $\rho, \eta \in \mathcal{T}_H$, the pair $\{\rho \wedge \eta, \rho \vee \eta\}$ equals $\{\rho, \eta\}$ pathwise. Since $\rho, \eta \leq H$, this gives $\mathbb{1}_{\{\rho \wedge \eta = H\}} + \mathbb{1}_{\{\rho \vee \eta = H\}} = \mathbb{1}_{\{\rho = H\}} + \mathbb{1}_{\{\eta = H\}}$. Moreover, $c(\rho \wedge \eta) + c(\rho \vee \eta) = c(\rho) + c(\eta)$, and hence $J_c(\rho \wedge \eta) + J_c(\rho \vee \eta) = J_c(\rho) + J_c(\eta)$. If $\rho, \eta \in \mathcal{S}(c)$, each term on the left is at most $\mathcal{V}(c)$, while their sum equals $2\mathcal{V}(c)$. Thus, both $\rho \wedge \eta$ and $\rho \vee \eta$ belong to $\mathcal{S}(c)$.

Claim C.4. *The set $\mathcal{S}(c)$ contains its essential infimum and essential supremum, which are its least and greatest elements.*

Because $\mathcal{S}(c)$ is downward directed, there is a decreasing sequence $(\sigma_n)_{n \geq 1} \subseteq \mathcal{S}(c)$ such that $\sigma_n \downarrow \rho_c^- := \text{ess inf}_{\rho \in \mathcal{S}(c)} \rho$. The filtration $\mathbb{F}^P$ is right-continuous, so ρ_c^- is a stopping time. Moreover, $\rho_c^- \leq \sigma_1$, so $\mathbb{E}[\rho_c^-] < \infty$. Since every optimizer has finite expected cost and $c(\sigma_n) \leq c(\sigma_1)$, the dominated convergence theorem gives $\mathbb{E}[c(\sigma_n)] \rightarrow \mathbb{E}[c(\rho_c^-)]$. Because $\sigma_n \leq H$, $\mathbb{1}_{\{\sigma_n = H\}} \rightarrow \mathbb{1}_{\{\rho_c^- = H\}}$ almost surely. Therefore, $J_c(\rho_c^-) = \mathcal{V}(c)$, so $\rho_c^- \in \mathcal{S}(c)$.

Likewise, upward directedness gives an increasing sequence $(\eta_n)_{n \geq 1} \subseteq \mathcal{S}(c)$ such that $\eta_n \uparrow \rho_c^+ := \text{ess sup}_{\rho \in \mathcal{S}(c)} \rho$. Optimality gives $\mathbb{E}[c(\eta_n)] = \mathbb{P}(\eta_n = H) - \mathcal{V}(c) \leq 1$. The monotone convergence theorem and the eventual linear lower bound on c imply $\mathbb{E}[\rho_c^+] < \infty$. Moreover, $\mathbb{1}_{\{\eta_n = H\}} \uparrow \mathbb{1}_{\bigcup_n \{\eta_n = H\}} \leq \mathbb{1}_{\{\rho_c^+ = H\}}$. Hence, $J_c(\rho_c^+) \geq \mathcal{V}(c)$. Feasibility gives the reverse inequality, so $\rho_c^+ \in \mathcal{S}(c)$. By construction, ρ_c^- and ρ_c^+ are respectively the least and greatest elements of $\mathcal{S}(c)$. $\blacksquare$

C.3 Proof of Proposition 5.5

Proof of Proposition 5.5. Let $\Delta := c_2 - c_1$. Since Δ is nondecreasing and $\Delta(0) = 0$, $c_1 \leq c_2$. Fix $\rho_1 \in \mathcal{S}(c_1)$ and $\rho_2 \in \mathcal{S}(c_2)$, and set $m := \rho_1 \wedge \rho_2$ and $j := \rho_1 \vee \rho_2$. Every optimizer has finite expected cost. Since $c_1 \leq c_2$, $\mathbb{E}[c_1(\rho_2)] < \infty$. Moreover, $c_2(m) \leq c_2(\rho_2)$ and $c_1(j) \leq c_1(\rho_1) + c_1(\rho_2)$, so all payoff differences below are finite.

Because $m \leq \rho_2$, monotonicity of Δ and optimality of ρ_2 under c_2 give

$$J_{c_1}(\rho_2) - J_{c_1}(m) = J_{c_2}(\rho_2) - J_{c_2}(m) + \mathbb{E}[\Delta(\rho_2) - \Delta(m)] \geq 0.$$

The modular identity established in [Lemma 5.2](#), applied under c_1 , yields $J_{c_1}(j) - J_{c_1}(\rho_1) = J_{c_1}(\rho_2) - J_{c_1}(m) \geq 0$. Since ρ_1 is c_1 -optimal, the left-hand side cannot be positive. Hence, equality holds and $j \in \mathcal{S}(c_1)$. Moreover, $J_{c_1}(\rho_2) - J_{c_1}(m)$ is the sum of two nonnegative terms and equals zero, so $J_{c_2}(\rho_2) = J_{c_2}(m)$. Thus, $m \in \mathcal{S}(c_2)$, proving the strong set order.

Apply the strong set order to $\rho_{c_1}^-$ and $\rho_{c_2}^-$. Their minimum belongs to $\mathcal{S}(c_2)$, and minimality of $\rho_{c_2}^-$ gives $\rho_{c_2}^- \leq \rho_{c_1}^- \wedge \rho_{c_2}^- \leq \rho_{c_2}^-$. Therefore, $\rho_{c_2}^- \leq \rho_{c_1}^-$. The strong set order also gives $\rho_{c_1}^+ \vee \rho_{c_2}^+ \in \mathcal{S}(c_1)$. Since $\rho_{c_1}^+$ is the greatest c_1 -optimizer, $\rho_{c_1}^+ \leq \rho_{c_1}^+ \vee \rho_{c_2}^+ \leq \rho_{c_1}^+$, and therefore $\rho_{c_2}^+ \leq \rho_{c_1}^+$. The persuasion-probability inequalities follow because $\{\rho = H\}$ is increasing in ρ on $\mathcal{T}_H$. $\blacksquare$

C.4 Proof of [Proposition 5.6](#)

Proof of [Proposition 5.6](#). Set $a := \frac{1}{\kappa_1^2}$ and $b := \frac{1}{\kappa_2^2}$, so $a \geq b > 0$. The cost difference is $d_{\kappa_1}(u) - d_{\kappa_2}(u) = c(au) - c(bu)$. A finite convex function is locally absolutely continuous, and its right derivative c'_+ is nonnegative and nondecreasing. For $v > u$,

$$[c(av) - c(bv)] - [c(au) - c(bu)] = \int_u^v [ac'_+(as) - bc'_+(bs)] ds \geq 0,$$

because $as \geq bs$, $c'_+(as) \geq c'_+(bs) \geq 0$, and $a \geq b$. Hence, $d_{\kappa_1} - d_{\kappa_2}$ is nondecreasing.

Apply [Proposition 5.5](#) to the common process P^1 , with d_{κ_2} as the lower cost and d_{κ_1} as the higher cost. This yields the strong set order comparison and the inequalities for the extremal stopping rules. The persuasion-probability inequalities follow from monotonicity of $\{U = H_1\}$ in U on $\mathcal{T}_{H_1}^1$. $\blacksquare$